Year Up United
Data Analyst Training Academy: Intro to Data Analysis & Excel Fundamentals

Week 1 Lab Workbook

This courseware is to be distributed to accompany instructor led training.

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1.A.1 Table of Contents

MODULE 1 GETTING STARTED 3

EXERCISE 1.A GETTING SET UP 4

MODULE 2 DATA ANALYSIS BASICS 5

EXERCISE 2.A THINKING CRITICALLY ABOUT DATA ANALYSIS 6
EXERCISE 2.B PERSONAL DATA ROUND-UP 10

MODULE 3 GATHERING DATA 12

EXERCISE 3.A REVIEWING SURVEY RESULTS 13
EXERCISE 3.B EXPLORING PUBLIC DATA 15
EXERCISE 3.C WRITING A SURVEY 17

MODULE 4 EXCEL FUNDAMENTALS 20

EXERCISE 4.A EXPLORING DATA WITH EXCEL 21
EXERCISE 4.B CLEANING AND SHAPING DATA WITH EXCEL 26
EXERCISE 4.C TRENDS DATA FOR MARKETING INSIGHTS 31

MODULE 5 DATA STORYTELLING 45

EXERCISE 5.A TELLING DATA STORIES WITH COMICS 46
EXERCISE 5.B RESTAURANT CUSTOMER SURVEY RESULTS 47

Module 1

Getting Started

Exercise 1.A Getting Set Up

Survey: How are you setting yourself up for success?

Use the survey link provided by your instructor to respond to each item below. (Note: Anonymized survey responses will be shared with the class.)

1. As a rule of thumb, getting to work or class on time means being *ready* for work/class at least a few minutes before it starts. Some people find this easy, but those of us who struggle with time management often leave getting ready to the last minute, which then turns into running late. To set yourself up for success in this class, it is helpful to build up a consistent routine of how you get started in the morning. **What do you envision yourself doing <u>5 minutes before</u> class starts each day, to ensure that you are ready to start on time?**

2. **Do you have a second monitor connected to your computer?** This could be an actual monitor or a TV. (Note that just having Zoom open on a second device like a tablet is not by itself a great option because it limits your ability to share your screen, which is really important for troubleshooting.) **If you don't have a second monitor, can you think of any friends or family members who you could ask to loan you one for the next 12 weeks?**

3. **Do you have an external keyboard and mouse?** (This isn't absolutely necessary, but it definitely helps!)

4. How do you plan to take notes for this class – in a doc on your computer, with pen and paper, or something else?

5. Practicing curiosity: What is one question that you have right now, about this course generally or about a specific topic we'll be talking about?

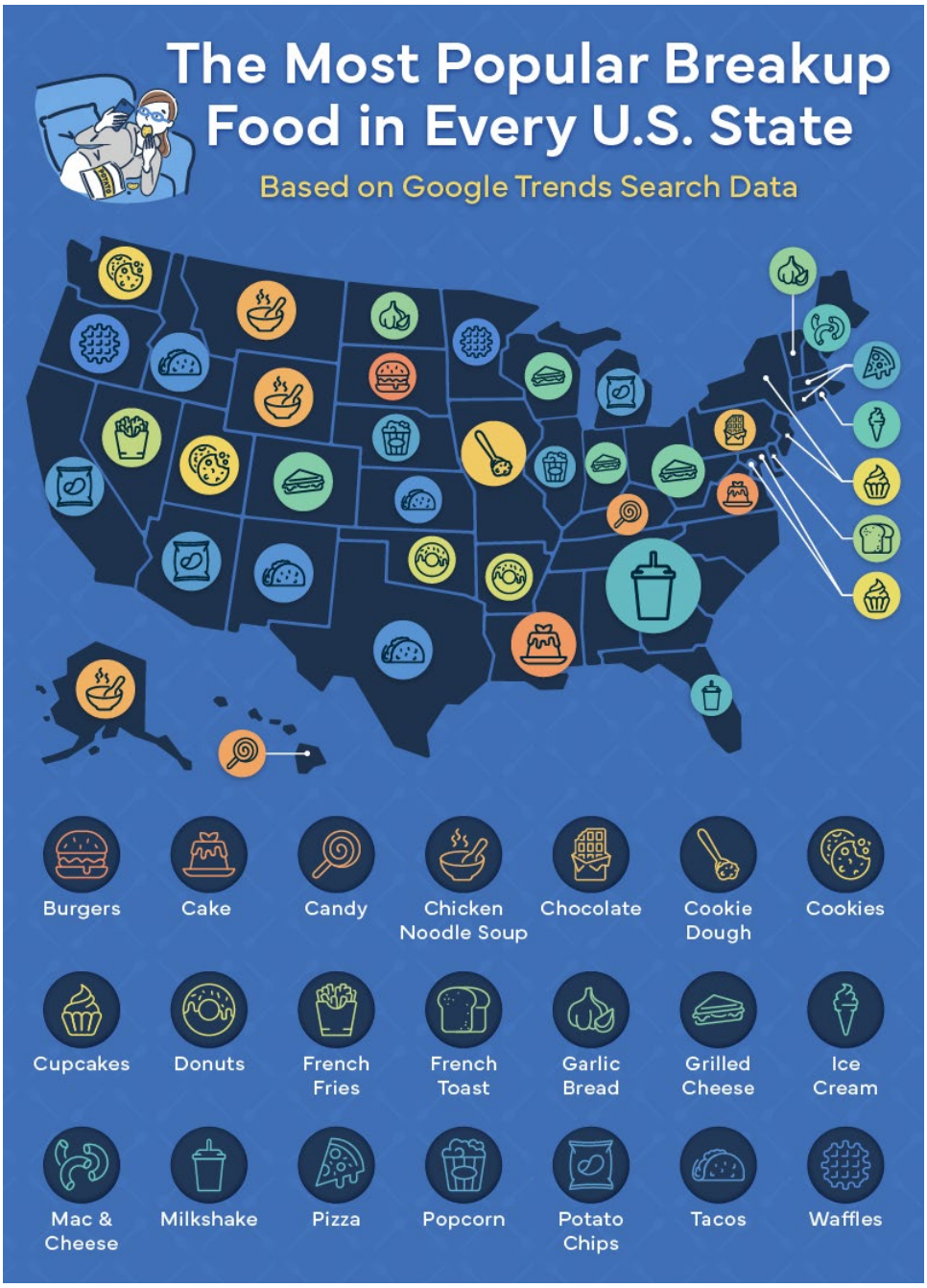
Module 2

Data Analysis Basics

Exercise 2.A Thinking Critically About Data Analysis

Exercising Your Critical Mind

Please take a few minutes to review the infographic from the article [The Most Popular Breakup Food in Every State](#) (you may also want to skim-read the article text):



At the bottom of the article, the author describes their methodology for compiling and analyzing this data: *To determine the most popular breakup comfort food in every U.S.*

state, we compiled a list of common post-breakup foods from sources like Bustle, The Recipe, and Beanstalk Mums. We then analyzed regional interest in these foods using Google Trends search data from February 2024 to February 2025. Finally, we compared search interest to find the most popular breakup food in each state. We used high search interest as an indicator of preference, highlighting the go-to comfort foods Americans turn to after heartbreak.

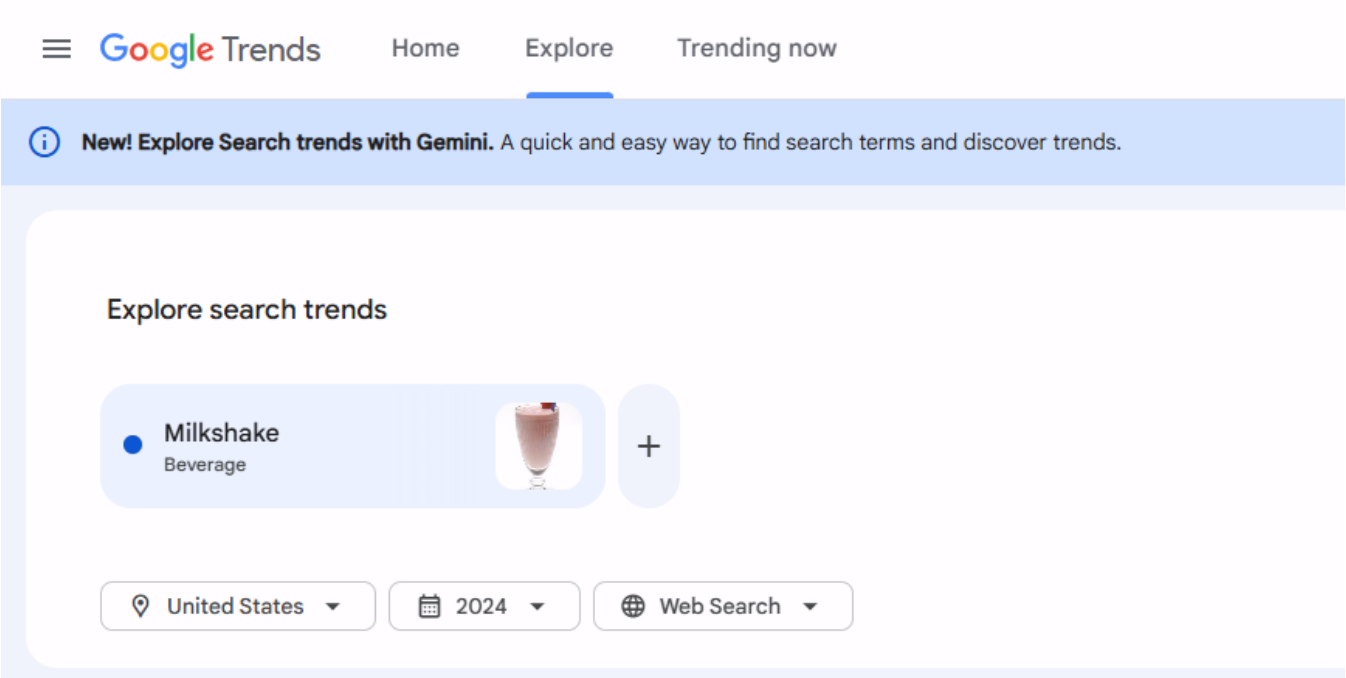
Get started by creating a new .txt file in the DataAnalytics/week-01 folder that you set up during pre-work. Name the file **Ex2A_ThinkingCritically.txt**

Open the .txt file in Notepad or other editor and write in your responses to the below questions. Copy and paste the full text of each question into the .txt file so that your answers have clear context:

1. Before doing any other investigation, list two or three questions that this infographic and explanation of methodology raise in your mind.

2. Where can you find Google Trends search data? Provide the URL.

3. The article concluded that "milkshakes take the crown as the ultimate breakup food, topping the list in six states." Pull up Google Trends data on Milkshake (beverage) and change the date range to February 1, 2024 through February 28, 2025.



   a) What is the highest peak recorded in the line chart for "Interest over time"?

   b) What is the lowest dip recorded?

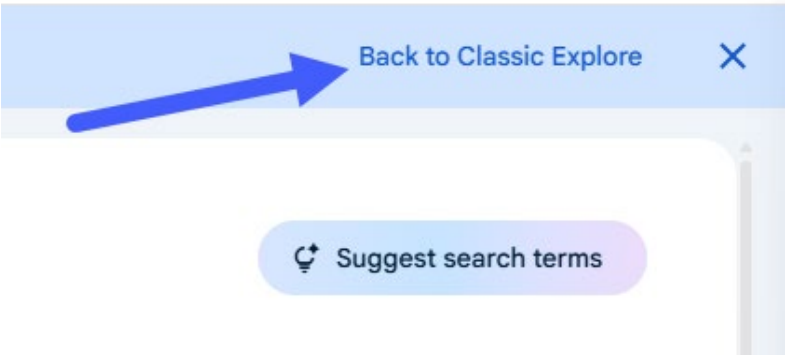
4. At the top next to the box for "Milkshake," there is a plus sign to add another search term. Click the plus sign to add a search for Hamburger (Food).
   a) What is the highest peak recorded for searches using Hamburger? What is the lowest dip?
   b) What happened to the line for Milkshake? What do the high and low look like now?

5. Why did the line for Milkshake change? Hint: Click on the ⓘ icon next to "Interest over time" for more information about how trend results are displayed.

6. Add a third term to your search, this time for Breakup (Topic).
   a) Do you see any patterns that suggest a relationship between searches for Breakup and Hamburger?
   b) How about between Breakup and Milkshake?

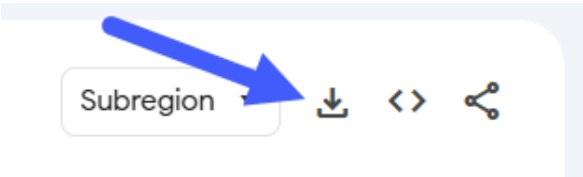
7. What does all of this suggest about the reliability of the conclusions reached in the article?

8. At the upper right of Google Trends, click on the link for "Back to Classic Explore."



   Notice that this view includes a section for "Compared breakdown by subregion."
   a) What was the top state by interest for Milkshake?
   b) What was the top state by interest for Hamburger?

9. Use the download icon to download a csv file with these search trend results. Save it to your DataAnalytics/week-01 folder as **GTrends_milkshake_hamburger_2024.csv**



10. OPTIONAL: Spend a little more time exploring Google Trends results for other "breakup foods" listed in the article. Some more food for thought:

    a) What are the differences you notice between searching for between searching for a term as a "search term" versus as a categorical term? What thoughts does this raise in your mind about working with Google Trends data?

    * For example, compare "Milkshake (beverage)" to "Milkshake (search term)", then compare "Hamburger (Food)" to "Burger (search term)"

    a) Try comparing "Hamburger (Food)" to some search terms oriented toward pop culture or current events. For instance, you might compare to "football" or "olympics." What do you notice? What do you think other patterns in search behavior might represent?

2. Save your updates to Ex2A_ThinkingCritically.txt.

3. Using Git Bash, commit and push your changes to the DataAnalytics repository.

    a) Remember the commands to add, commit and push:

```
git add .

git commit -m "Added Exercise 2A docs"

git push
```

4. As a final check, go to your GitHub profile, click on DataAnalytics, and confirm that your homework assignment appears in DataAnalytics/week-01 as expected.

Exercise 2.B Personal Data Round-up

1. This exercise has you consider sources of data that are available to you *about yourself*. To begin, open a blank Excel spreadsheet. Use File > Save As to save your spreadsheet to DataAnalytics/week-01, with the filename **Ex2B_data_roundup.xlsx**

2. In the first row, enter the following column names:

   a) Name of data source

   b) Type of info (e.g. social media, health, communication, purchases, services)

   c) Depth of coverage (i.e. How far back does the data go?)

   d) Downloadable?

   e) Steps to download

   f) Download format (e.g. .csv, .json, .pdf, etc)

   g) Additional notes (optional, if you want to jot down any observations)

   Note: *Do <u>not</u>* include any login or other private personal info in the spreadsheet!

3. For each source that you can think of, fill in a new row. You do not need to download the actual data, but walk through the steps to figure out how you could. Example:

Steps to download
Settings > Accounts Center > Your information and permissions > Download your information
Settings > Accounts Center > Your information and permissions > Download your information
Menu > Sharing > Computer-Readable Export
Account Settings > Export Data

4. Complete this process for 3 to 6 data sources.

5. Make sure your changes to Ex2B_data_roundup.xlsx are saved. (Excel turns on auto-save by default, but you should still always confirm that changes are saved!)

6. Using Git Bash, commit and push your changes to the DataAnalytics repository.

7. As a final check, go to your GitHub profile, click on DataAnalytics, and confirm that your homework assignment appears in DataAnalytics/week-01 as expected. (Notice that with an Excel file, you will be able to see that the file is saved and download the file to review it, but you will *not* be able to view a preview of the file online.)

Module 3

Gathering Data

Exercise 3.A Reviewing survey results

Getting Familiar with Surveys and Built-In Analytics

Use the link provided by your instructor to access your class's full responses to the survey from Exercise 1.A, Getting Set Up.

Get started by creating a new .txt file in the DataAnalytics/week-01 folder named **Ex3A_ReviewingSurveyResults.txt**

Open the .txt file in Notepad or other editor and write in your responses to the below questions. Copy and paste the full text of each question into the .txt file so that your answers have clear context:

Click on "View responses" to see the Responses Summary. In your .txt file, write numbers 1 through 5 to represent the five survey questions. For **each** survey question, consider the following:

1. What conclusion can you reach about the responses to this question?

2. Is the default summary or visualization helpful? How could it be improved?

3. For questions with an "Other" option, where can you find the actual text of those responses? Do you notice anything that might be tricky about analyzing these?

4. For questions calling for a written response, what challenges do you notice about finding trends or patterns in the answers provided?

5. Now that you can see the responses, how might you recommend rephrasing this or question or the response options in a future survey?

Next respond to the below questions:

6. Is there anything that surprised you about the survey responses?

7. Looking at the responses to questions 1 and 5, are there any ideas you might try to include in your morning routine, or any questions that you are also curious about?

8. Pretend you are the instructor. Is there anything you can learn from this survey to help in how you approach teaching this class?

9. Again, pretend you are the instructor. Do you think it was helpful or not helpful to make this survey anonymous? Why or why not? (If you can see both pros and cons, you may include both.)

On the Responses Overview page for the form, under “Insights and actions”, click on the button to open the survey responses in Excel. Take a moment to review the layout of the data. Then at the top click on File > Create a Copy > Download a Copy to download the responses to your DataAnalytics/week-01 folder as **Exercise 1.A Getting Set Up.xlsx**

When you are finished, make sure your answers to Ex3A_ReviewingSurveyResults.txt are saved.

Then using Git Bash, commit and push your changes to the DataAnalytics repository.

Exercise 3.B Exploring public data

Using Data.gov

Get started by creating a new .txt file in the DataAnalytics/week-01 folder named **Ex3B_ExploringPublicData.txt**

Open the .txt file in Notepad or other editor and write in your responses to the below questions. Copy and paste the full text of each question into the .txt file so that your answers have clear context:

1. Start by reviewing the data.gov home page. What is this?

2. Go to **Most Viewed Datasets**. How many datasets are listed?

3. Find the **Topic Category** for Human Health. Review the range of data sources available for this topic. Why do you think these were tagged with the Human Health category?

4. Go to the **Food Price Outlook**. Review the catalog page for this dataset. How can you navigate to additional datasets from the USDA?

5. Go back to the data.gov home page. This time, click on **Datasets by Organization**.

6. Try searching for your city, county, and state. What organizations are returned?

7. Search for USDA, and click into the result. Within the USDA datasets, search for **food composition**. Notice that the top three results are all datasets concerning nutrition information. Try searching the USDA page for **nutrition**. How does that affect your results?

   a) One of the tricky things about working with data is that sometimes the terminology used by a data source is not obvious. You may need to dig around and be creative to find what you're looking for.

8. Go back to the results for **food composition**, and click on **FoodData Central**. Review the catalog page for this data source, then click on the button for **Visit Page**. Take some time to explore the FoodData Central website.

a) Try searching for **butter**, or any of your favorite foods. What kind of results do you get? What happens when you click on a result?

b) Go back to the home page. Find **Foundation Foods**, and click on Downloads. What kind of data can you find on this page? How big are these files? Where can you find more information on the data types available?

9. When you are finished, make sure your changes to Ex3B_ExploringPublicData.txt are saved.

10. Using Git Bash, commit and push your changes to the DataAnalytics repository.

Exercise 3.C Writing a Survey

Scenario

Imagine that you are at your actual-real-life favorite restaurant (or coffee shop, café, pub, or other similar establishment). It just so happens that one day while you're there, the manager strikes up a conversation with you and finds out that you're interested in data analytics. The manager says that they're looking to start collecting more direct customer feedback. In addition to general feedback, they're thinking of starting a loyalty program or offering new perks to frequent customers, and they would like some information to help decide what that program might look like.

The manager asks if you might be able to help them out in exchange for a nice gift card worth a few meals on the house. You say, no problem! So… how are you going to go about this?

1. Before diving into trying to build the actual survey form, spend a little time thinking and making notes about what kind of information you want to gather.
   a) What kind of customer data will be useful to the restaurant? How will they want to use that customer information?
   b) You're a customer – what kind of feedback would *you* want to give? What things are unique to this restaurant that you might ask about?
   c) What kind of questions might yield useful information that could help shape the direction of the new customer reward program?
   d) How long should the survey be? Consider that typing text responses takes longer than choosing multiple choice options. How does that impact what questions you ask?

2. For this exercise, please use Microsoft Forms, https://forms.office.com/ and sign in with your Year Up United credentials.

3. For some inspiration, take a look at the templates available from Forms, especially the "Customer satisfaction survey" and "Market research survey."

4. Once you have explored the templates, go back to the Forms home screen, and click on **New Form** to start with a blank form.

5. Give your form a title and description.

   a) Clearly include the name of the restaurant plus a brief description of what they offer. Remember that this is customer-facing, so make it friendly and in keeping with the restaurant's brand.

6. Use the **Add New** button to begin adding questions to your survey.

   a) Review your notes, and start creating questions. Your survey should include 5 to 10 questions total.

   b) Consider including some questions about the customer's recent experience and some geared towards researching a possible rewards program. Include a variety of question types (e.g. multiple choice, text box, rating).

   c) There is no strict requirement for the number of questions, but avoid asking more than 10. Ideally the survey should take no more than five minutes to fill out – you want a good amount of information, but you don't want customers to get bored and abandon it halfway through!

7. Use the **Style** options to give an appealing format to your form.

8. Make sure to use **Preview** to see what your form will look like from the customer's point of view.

9. When you are happy with your survey, use the **Collect Responses** button to set up options for sharing.

   a) Consider what options might work best to share this survey with customers. How can you make it easy for customers to respond? What other questions might you need to include if you want to follow up with customers later?

10. Copy the link or QR code to share your survey, and post it to the class channel.

Turning the tables

11. We've now got a bunch of surveys that we need to generate sample responses for. Luckily, we also have a whole class full of people! Please start clicking into your classmates' shared surveys. Review the description provided, and imagine that you are a customer of this restaurant. Then reply to the questions as a customer might.
    a) You can choose how you respond to the questions. Are you a happy customer? Was there anything wrong with your order, or was everything great? Are you a frequent customer here, or is this your first time visiting?
    b) For any questions asking for email / phone number / address (or other personally identifiable information), please feel free to make up any information you like.
    c) If you need inspiration on how to respond, try checking out the restaurant's website or reviews on Google or Yelp.

12. As you take each survey, make notes for yourself with your feedback on how the survey was written.
    a) Were there any questions you particularly liked? Anything you thought was confusing?
    b) Was there something you wished you had come up with?
    c) Do you have any suggestions for how you might frame a question differently? Or follow-up questions that might be helpful to include?

Bonus Lab: Sharing the results

13. Once most of the class has had a chance to take your survey, take some time to review the results. As an optional **bonus lab**, record a short video to describe and share your survey results (and make sure to share your screen in the recording!).
    a) Start by briefly introducing the restaurant you chose.
    b) Explain what information you wanted to collect and your initial hypothesis for what results would be.
    c) Use the visualizations generated by Microsoft Forms, walk through the results, going through each question and sharing what you found interesting or surprising about the responses.

14. When you are done, post the link to your recording to the class Teams channel.

Module 4

Excel Fundamentals

Exercise 4.A Exploring data with Excel

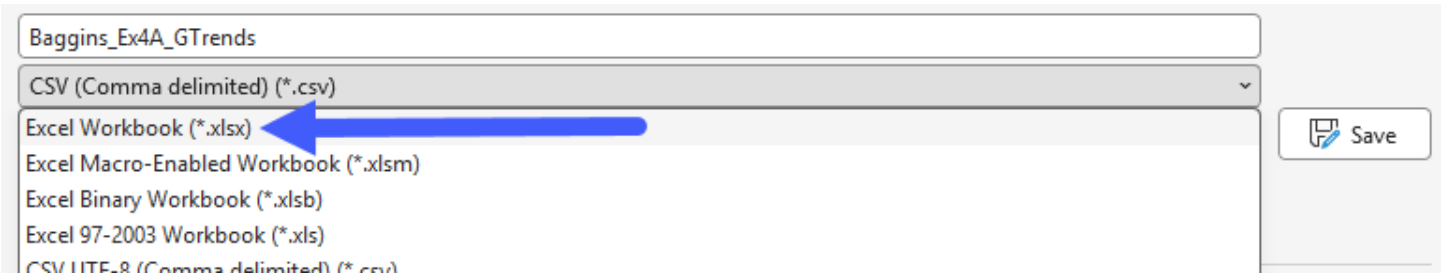
Working with Google Trends data

1. Open the file GTrends_milkshake_hamburger_2024.csv from where you previously saved it. Your laptop should default to opening the file with Excel, but if it does not, within File Explorer you can right-click the file, choose Open With, then select Excel.

   The first lines of the spreadsheet should look like this:

	A	B	C	D	E	F	G
1	Category: All categories						
2							
3	Region	Milkshake	Hamburge	Breakup: (2/1/24 - 2/28/25)			
4	South Car	22%	67%	11%			
5	North Car	24%	63%	13%			
6	Tennessee	21%	67%	12%			
7	Indiana	24%	63%	13%			
8	Kentucky	22%	65%	13%			
9	Alabama	23%	65%	12%			
10	West Virgi	24%	64%	12%			
11	Delaware	25%	62%	13%			

2. Use File > Save As to save your spreadsheet to DataAnalytics/week-01. Name the file **Ex4A_GTrends.xlsx** and change the file type to Excel Workbook (*.xlsx).



   Make sure that you change the file type! Otherwise, a lot of your work for this lab will be lost if it gets saved as a CSV file.

3. Resize columns A through D to fit the text.

	A	B	C	D
1	Category: All categories			
2				
3	Region	Milkshake: (2/1/24 - 2/28/25)	Hamburger: (2/1/24 - 2/28/25)	Breakup: (2/1/24 - 2/28/25)
4	South Carolina	22%	67%	11%
5	North Carolina	24%	63%	13%
6	Tennessee	21%	67%	12%
7	Indiana	24%	63%	13%
8	Kentucky	22%	65%	13%
9	Alabama	23%	65%	12%

4. The first row contains the text "Category: All categories." This is not useful. Delete the top two rows. The result should be that "Region" now appears in cell A1.

	A	B	C	D
1	Region	Milkshake: (2/1/24 - 2/28/25)	Hamburger: (2/1/24 - 2/28/25)	Breakup: (2/1/24 - 2/28/25)
2	South Carolina	22%	67%	11%
3	North Carolina	24%	63%	13%
4	Tennessee	21%	67%	12%
5	Indiana	24%	63%	13%
6	Kentucky	22%	65%	13%
7	Alabama	23%	65%	12%

5. Using the **Sort & Filter** options found in the ribbon, turn on filters. A drop-down option should appear for each column header.

	A	B	C	D
1	Region	Milkshake: (2/1/24 - 2/28/25	Hamburger: (2/1/24 - 2/28/25	Breakup: (2/1/24 - 2/28/25
2	South Carolina	22%	67%	11%
3	North Carolina	24%	63%	13%
4	Tennessee	21%	67%	12%
5	Indiana	24%	63%	13%
6	Kentucky	22%	65%	13%
7	Alabama	23%	65%	12%

a) Click the drop-down for Region and select Sort A to Z to sort the list alphabetically.

6. How many Regions are in the list in Column A?

a) Add the label "# of regions" to cell F1. If the text in cell F1 is cut off, resize the column so that the full text is visible.

b) Then calculate the answer by adding the following formula to cell G1:
`=COUNTA(A2:A52)`

c) Use the formatting options in the ribbon to change the fill color for F1 and G1 to light blue.

7. Now use **Freeze Panes** to make the headings display persistently, even when you scroll down the list.

   a) Go to the View tab of the ribbon.

   b) Click the button for Freeze Panes.

   c) Click the option Freeze Top Row. Notice that the border under the top row subtly changes from a gray line to a darker line, providing a visual hint that Freeze Panes is turned on.

   d) Try scrolling down vertically to test out how the frozen headers look.

8. In this spreadsheet, the highest percentage values belong to the Hamburger column, but it is the second search term listed. Move the Hamburger column in front of the column for Milkshake.

   a) Click on the C at the top of the Hamburger column to highlight the entire column, then hit Ctrl+X (or right-click and hit Cut) to copy the column.

   b) Click on the B at the top of the Milkshake column, then right-click and select **Insert Cut Cells**.

   c) Your updated columns should look like this:

	A	B	C	D	E	F	G
1	Region	Hamburger: (2/1/24 - 2/28/25	Milkshake: (2/1/24 - 2/28/25	Breakup: (2/1/24 - 2/28/25		# of regions	51
2	Alabama	65%	23%	12%			
3	Alaska	72%	15%	13%			
4	Arizona	67%	20%	13%			
5	Arkansas	67%	21%	12%			
6	California	62%	20%	18%			
7	Colorado	68%	18%	14%			
8	Connecticut	65%	21%	14%			

9. Next insert a new column before Region.

   a) In the first row, write in the header name RegionID.

b) In the second and third rows, type the values 1 and 2.

	A	B	C	
1	RegionID	Region	Hamburger: (2/1/24 - 2/28/25	Milksha
2	1	Alabama	65%	
3	2	Alaska	72%	
4		Arizona	67%	
5		Arkansas	67%	
6		California	62%	
7		Colorado	68%	

c) With both cells containing 1 and 2 selected, hover your mouse over the little square in the bottom right corner so that your mouse converts into a plus symbol +:

	A	B	C	
1	RegionID	Region	Hamburger: (2/1/24 - 2/28/25	Milksha
2	1	Alabama	65%	
3	2	Alaska	72%	
4			67%	
5		Arkansas	67%	
6		California	62%	
7		Colorado	68%	

d) Double click with the + cursor to auto-fill the remaining rows with sequential numbers.

	A	B	C	
1	RegionID	Region	Hamburger: (2/1/24 - 2/28/25	Milksha
2	1	Alabama	65%	
3	2	Alaska	72%	
4	3	Arizona	67%	
5	4	Arkansas	67%	
6	5	California	62%	
7	6	Colorado	68%	

10. Notice that your new RegionID column does not automatically have a filter drop-down.

a) What happens when you use the filter drop-down for Region to change the sort order to Z to Z (reverse alphabetical)?

	A	B	C	
1	RegionID	Region	Hamburger: (2/1/24 - 2/28/25	Milksha
2	1	Wyoming	71%	
3	2	Wisconsin	66%	
4	3	West Virginia	64%	
5	4	Washington	67%	
6	5	Virginia	64%	
7	6	Vermont	68%	
8	7	Utah	65%	

The RegionID numbers stay in the same position, but now they line up with totally different states. Oops!

b) Click Undo (Ctrl+Z) or re-filter to revert the Region column back to A to Z sort order.

c) Then go to the Sort & Filter options in the ribbon to turn off column filters by clicking Filter to deselect it.

d) Finally, click Filter again to turn column filters back on.

e) Now columns A through E should *all* have filter drop-downs.

	A	B	C	D	E	F	G	H
1	Region	Region	Hamburger: (2/1/24 - 2/28/25	Milkshake: (2/1/24 - 2/28/25	Breakup: (2/1/24 - 2/28/25		# of regions	51
2	1	Alabama	65%	23%	12%			
3	2	Alaska	72%	15%	13%			
4	3	Arizona	67%	20%	13%			
5	4	Arkansas	67%	21%	12%			
6	5	California	62%	20%	18%			
7	6	Colorado	68%	18%	14%			
8	7	Connecticut	65%	21%	14%			

Notice, however, that columns G and H do *not* have filtering. This is because there is a blank column in between E and G. If you wanted to include a larger filterable area irrespective of blank columns, you would need to manually select the full desired area before turning on filters (e.g. by highlighting columns A through H).

11. Make sure your changes are saved to Ex4A_GTrends.xlsx

12. Using Git Bash, commit and push your changes to the DataAnalytics repository.

Exercise 4.B Cleaning and shaping data with Excel

Continuing to work with Google Trends data

1. Open Ex4A_GTrends.xlsx to use as a starting point.

2. Go to File > Save a Copy, and save to DataAnalytics/week-01 with the new filename **Ex4B_more_GTrends.xlsx**

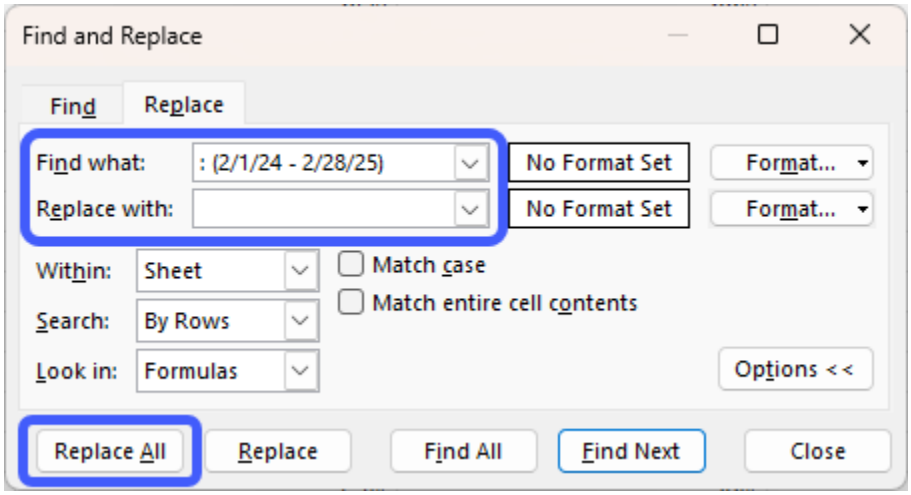
3. Notice that our search term column headers in C through E all contain that date range that was used in searching Google Trends. Let's relocate that information to one spot.

   a) In cell J1, write "date range" and adjust the column width so that the text is not cut off.

   b) Copy the date value 2/1/24 - 2/28/25 into cell K1. Again, adjust the column width so that the text is not cut off.

   c) Change the fill color for J1 and K1 to light gray.

   d) Now go to the ribbon and click on Find & Select > **Replace** (or hit Ctrl+H). In the Find and Replace pop-up dialog, in the "Find what" box enter the full text that to be removed from the column headers, starting with the colon and ending with the closed parenthesis. Leave "Replace with" blank. Then click Replace All:



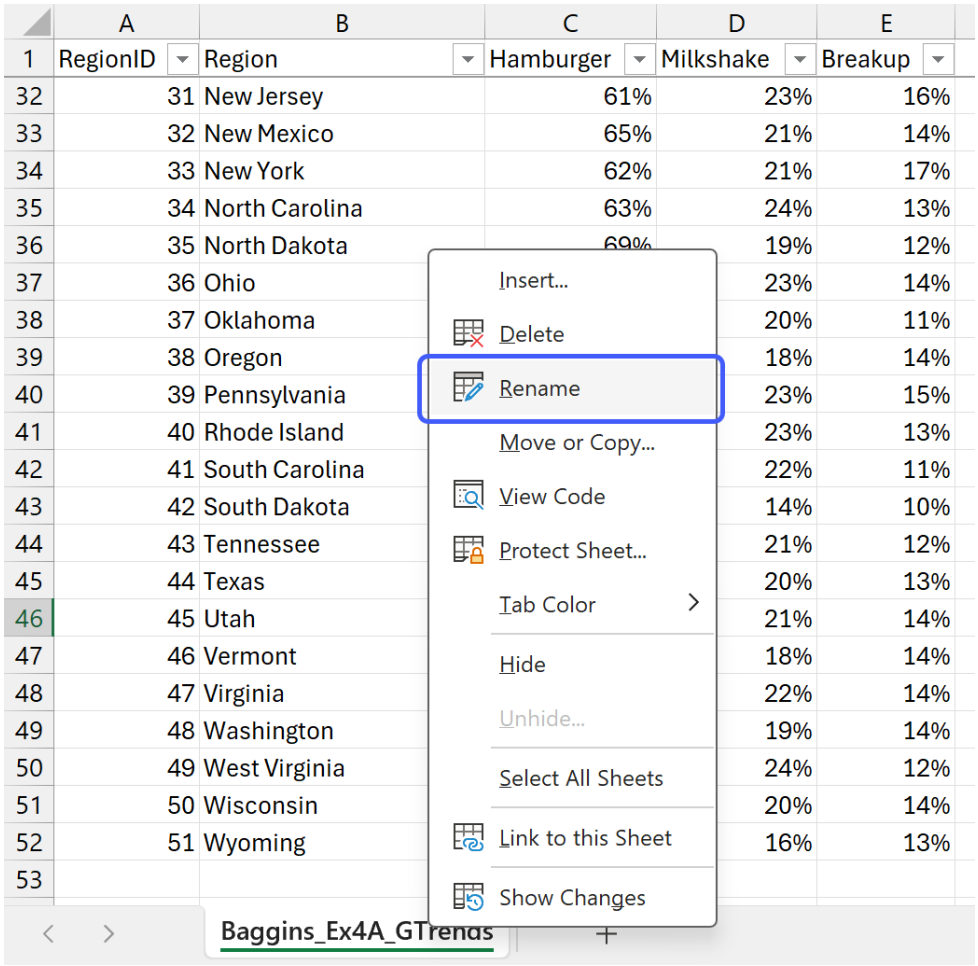
   You should receive a pop up saying you successfully made 3 replacements. Close the pop-up, then close the Find & Replace dialog.

e) Resize columns C through E remove excess column width.

	A	B	C	D	E	F	G	H	I	J	K
1	RegionID	Region	Hamburger	Milkshake	Breakup		# of regions	51		date range	2/1/24 - 2/28/25
2	1	Alabama	65%	23%	12%						
3	2	Alaska	72%	15%	13%						
4	3	Arizona	67%	20%	13%						
5	4	Arkansas	67%	21%	12%						
6	5	California	62%	20%	18%						
7	6	Colorado	68%	18%	14%						

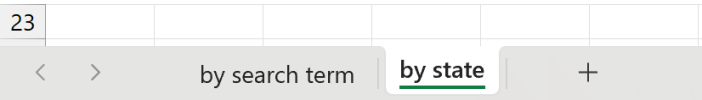
4. The spreadsheet currently is working with data that is tall and skinny – lots of rows with few columns. But what if we wanted to change how we're looking at the data? Let's add a second tab to this workbook and experiment!

a) At the bottom of the screen, right-click on the first and only tab of your workbook (named after the filename you started from, Ex4A_GTrends), and select Rename.



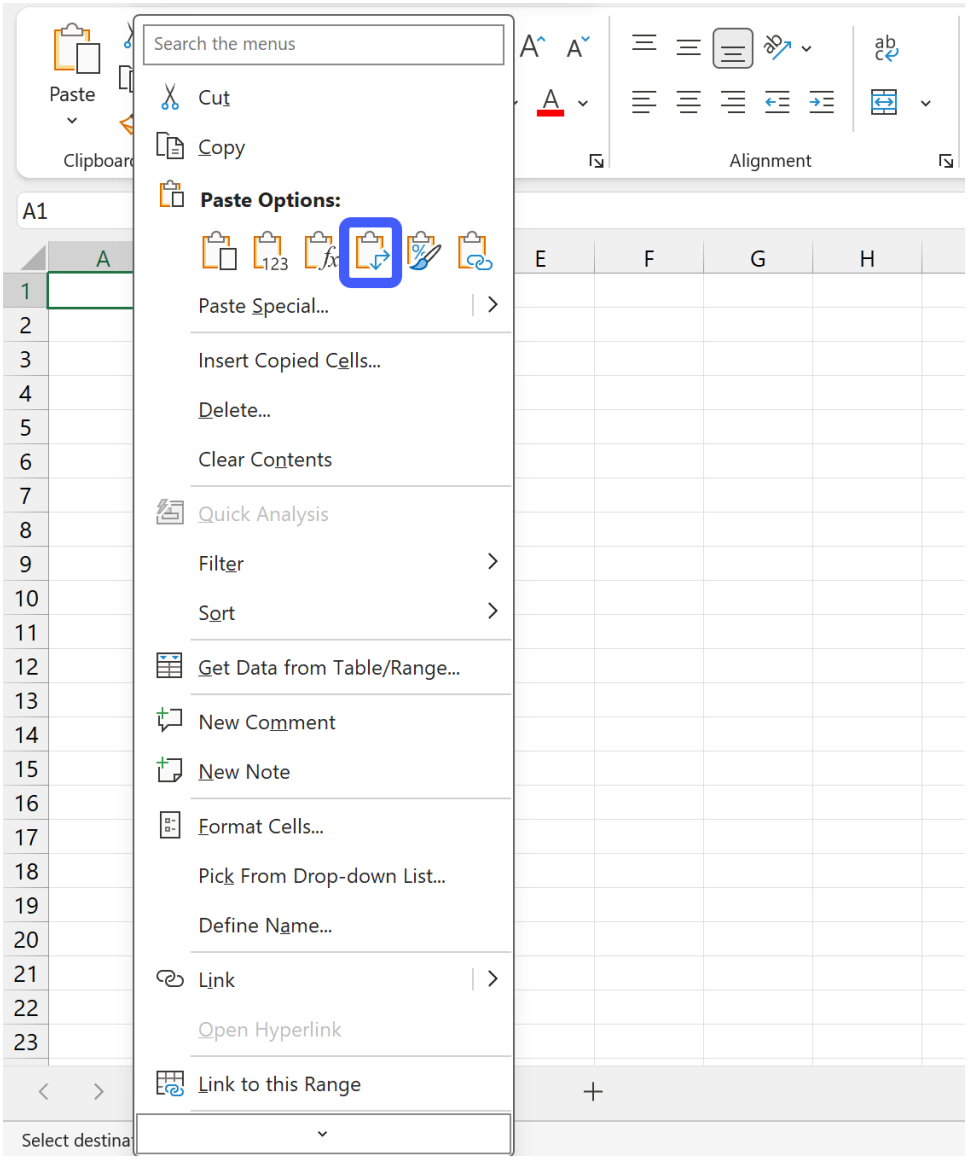
Change the tab name to **by search term**.

b) Click the + symbol next to the first tab to add a new tab. Then rename the new tab to **by state**.



c) In the “by search term” tab, select the cells in columns B through E, from row 1 through row 52. (You can do this by clicking and dragging with your mouse, or by selecting cell B1 at the top, then holding shift and using the arrow keys on your keyboard to select through column E and row 52.) When you have selected the full area B1:E52, hit Ctrl+C (or right-click and select Copy)

d) In the “by state” tab, select the first cell, A1. Right-click on A1 and select the transpose icon:



e) Your end result should be four rows tall and many columns wide:

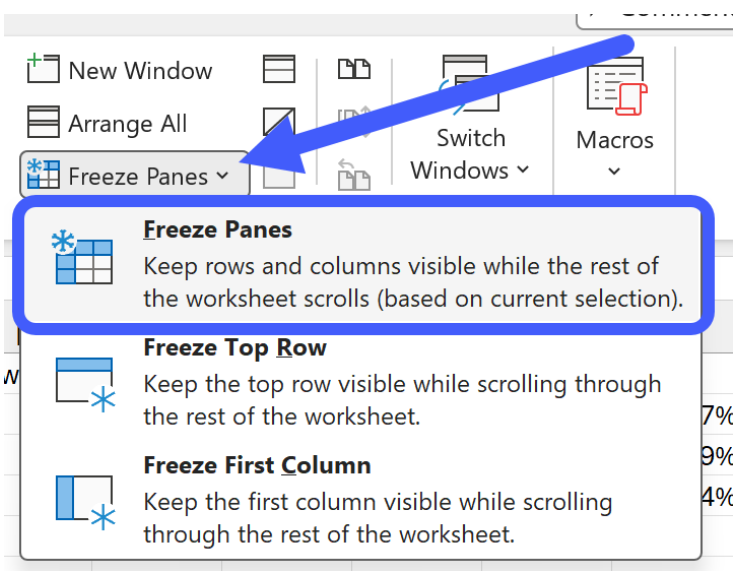
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Region	Alabama	Alaska	Arizona	Arkansas	California	Colorado	Connectic	Delaware	District of	Florida	Georgia	Hawaii
2	Hamburge	65%	72%	67%	67%	62%	68%	65%	62%	67%	64%	62%	73%
3	Milkshake	23%	15%	20%	21%	20%	18%	21%	25%	17%	22%	24%	14%
4	Breakup	12%	13%	13%	12%	18%	14%	14%	13%	16%	14%	14%	13%
5													

5. Right now this is a very short list, but imagine you are just starting a project in which you will be adding a hundred more search terms, making your data both tall *and* wide. How might you use the Freeze Panes feature to make it easier to review this data?

a) You can set a custom option to freeze all rows above and all columns to the left of a selected cell. To see this in action, first select cell B2 in the current sheet.

	A	B	C	D	E	F	G
1	Region	Alabama	Alaska	Arizona	Arkansas	California	Colorado
2	Hamburge	65%	72%	67%	67%	62%	68%
3	Milkshake	23%	15%	20%	21%	20%	18%
4	Breakup	12%	13%	13%	12%	18%	14%
5							

b) Go to the View tab of the ribbon, click the button for Freeze Panes, then click the menu option for Freeze Panes.



As before, notice the darkened borders beneath the top row and to the right of the first column that indicate Freeze Panes is turned on:

	A	B	C	D	E	F	G
1	Region	Alabama	Alaska	Arizona	Arkansas	California	Colorado
2	Hamburge	65%	72%	67%	67%	62%	68%
3	Milkshake	23%	15%	20%	21%	20%	18%
4	Breakup	12%	13%	13%	12%	18%	14%
5							

Experiment with scrolling horizontally and vertically to see the full effect.

6. Make sure your changes to Ex4B_more_GTrends.xlsx are saved.

7. Using Git Bash, commit and push your changes to the DataAnalytics repository.

Exercise 4.C Trends data for marketing insights

Working with a scenario

For this exercise, you will continue working with the Google Trends data where you left off previously. Open Ex4B_more_GTrends.xlsx and save a copy with the new name **Ex4C_GTrends_marketing.xlsx**

Imagine you are an intern in the marketing department of a restaurant conglomerate that owns fast food and fast-casual restaurant chains. You've received an assignment from your supervisor: The team wants to know the relative popularity of searches for milkshakes and hamburgers for two different sales regions, the Northwest region and the North Central region.

* The Northwest sales region includes the states Washington, Oregon, Idaho, and Montana.
* The North Central sales region includes the states North Dakota, South Dakota, Nebraska, Minnesota, Wisconsin, and Michigan.

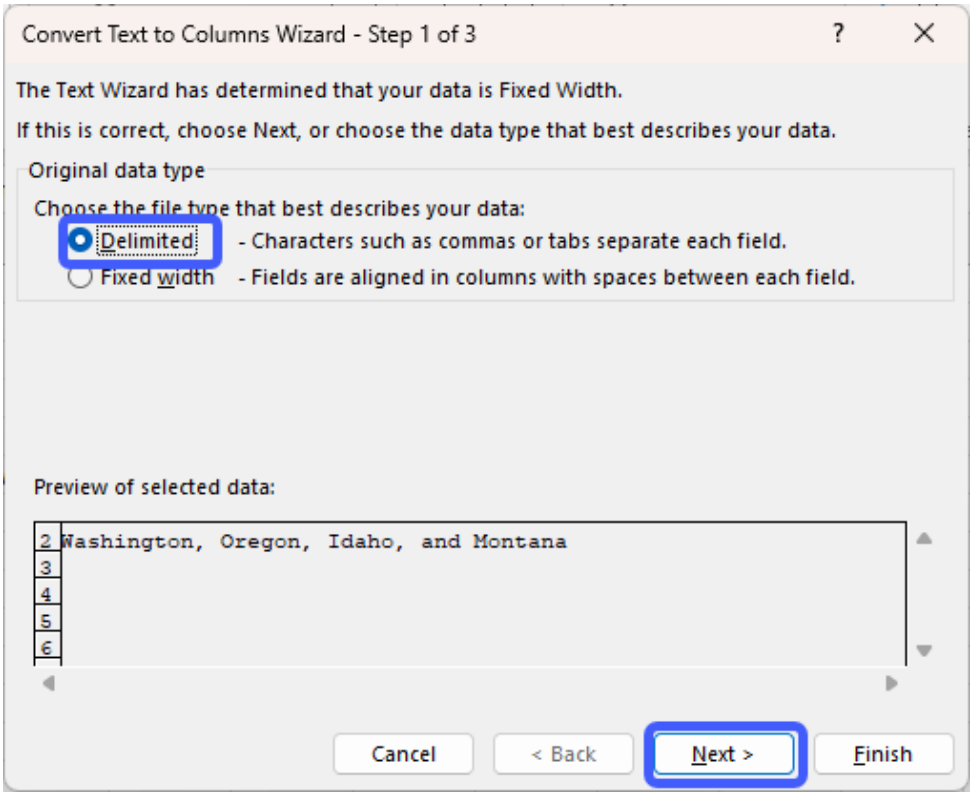
1. Add a third tab to your spreadsheet and name it **sales region lookup**.

2. Double-click on cell A1, then copy and paste the list of Northwest states (commas and all!).

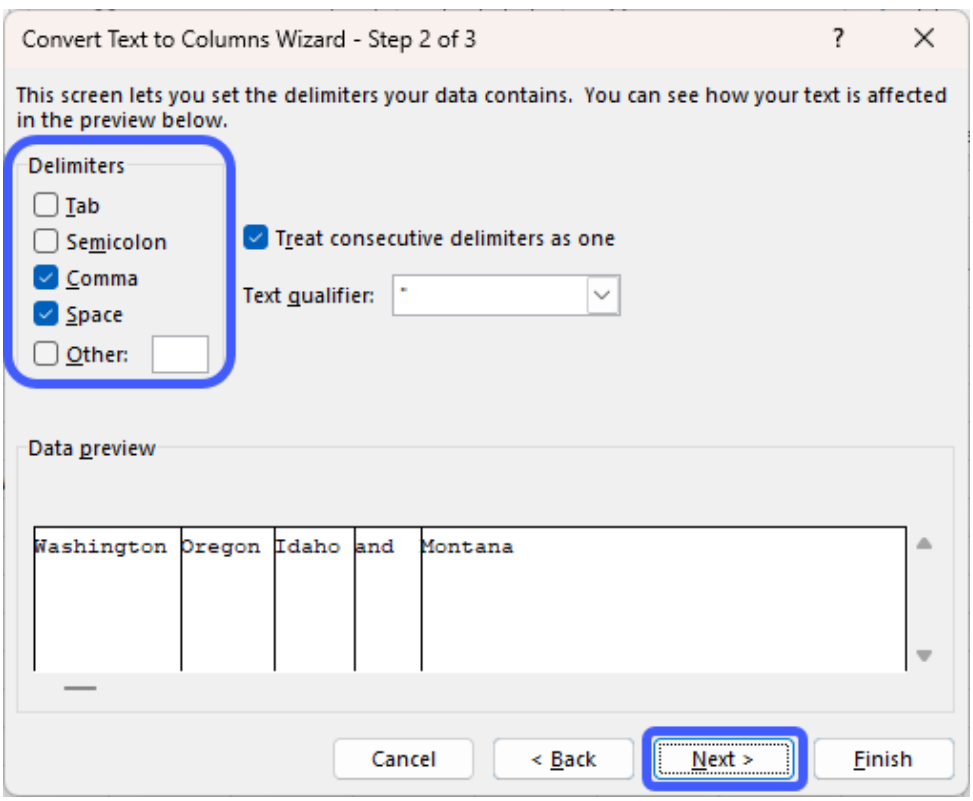
	A	B	C	D	E	F	G
1	Washington, Oregon, Idaho, and Montana						
2							

(Un-formatting tip: Double-clicking on the cell – or clicking into the formula bar to edit text there – ensures that you do not overwrite the cell's existing formatting. You can also right click and use Paste Special > Text, or Ctrl+Shift+V if enabled.)

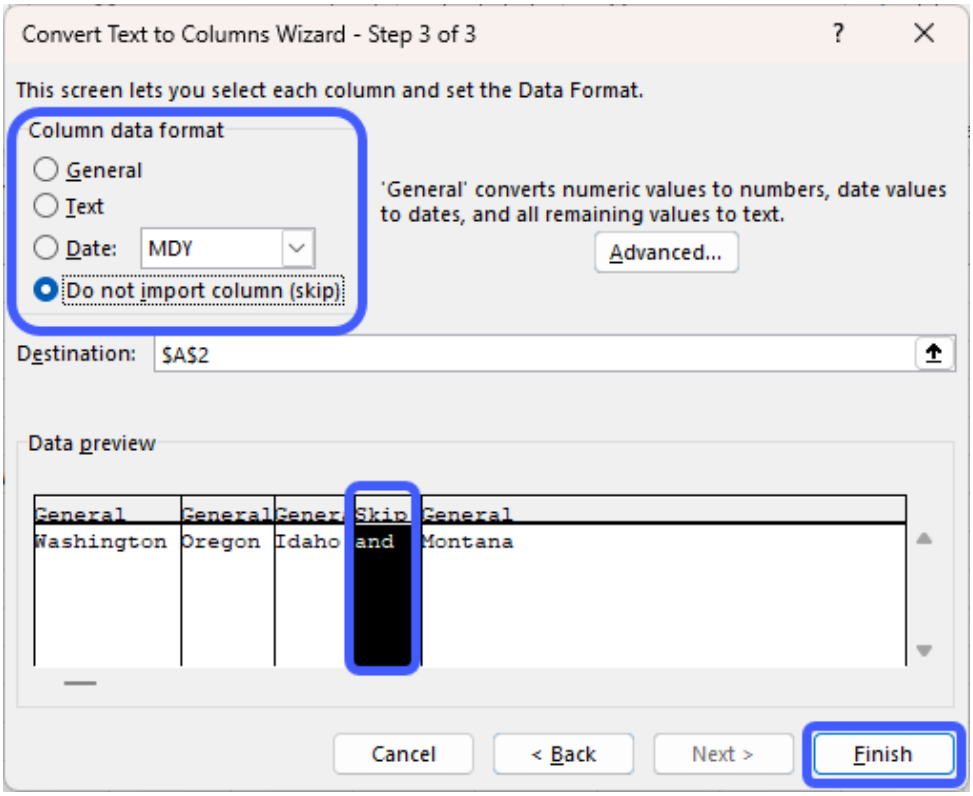
3. With cell A1 still selected, go to the Data tab of the ribbon, and click on **Text to Columns**. Select the option for Delimited, and click Next.



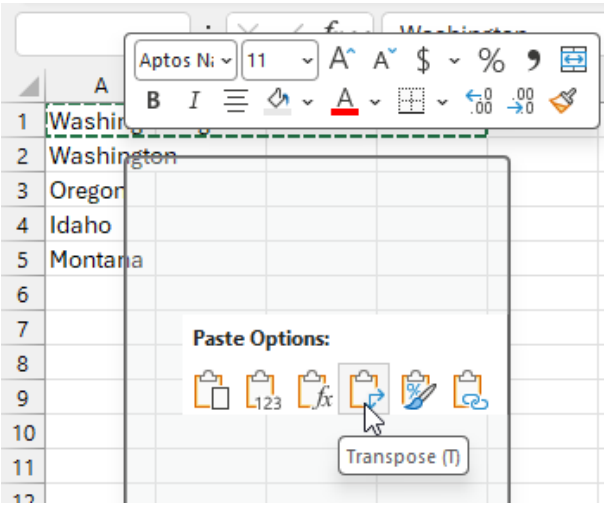
4. Under Delimiters, uncheck Tab. Add checks next to Comma and Space. Then click Next.



5. In the “Data preview” pane, notice that the output includes the word “and” which we don’t need. Click on the preview column where “and” appears. Then under “Column data format”, select “Do not import column (skip)”. Finally, click Finish.



6. Highlight your horizontal row of state names. Then right-click cell A2 and paste using Transpose.

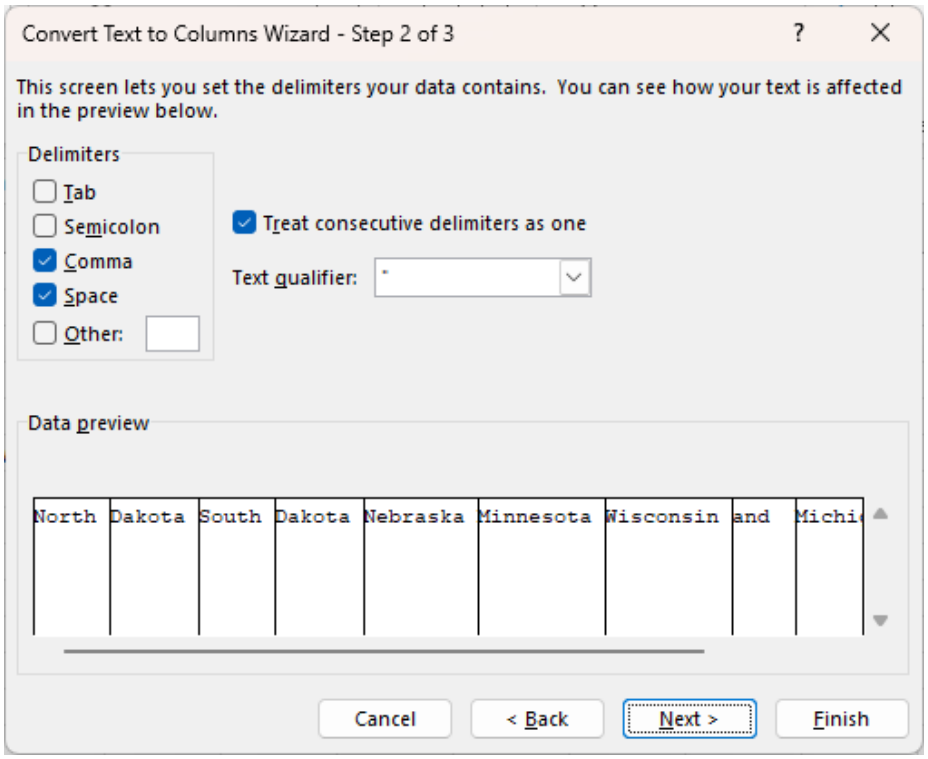


7. Clear the contents of row 1 by highlighting and clicking the delete key on your keyboard, or using right-click > Clear Contents.

6. Continuing with the internship scenario, you can follow similar steps to separate the values for the North Central states: North Dakota, South Dakota, Nebraska, Minnesota, Wisconsin, and Michigan

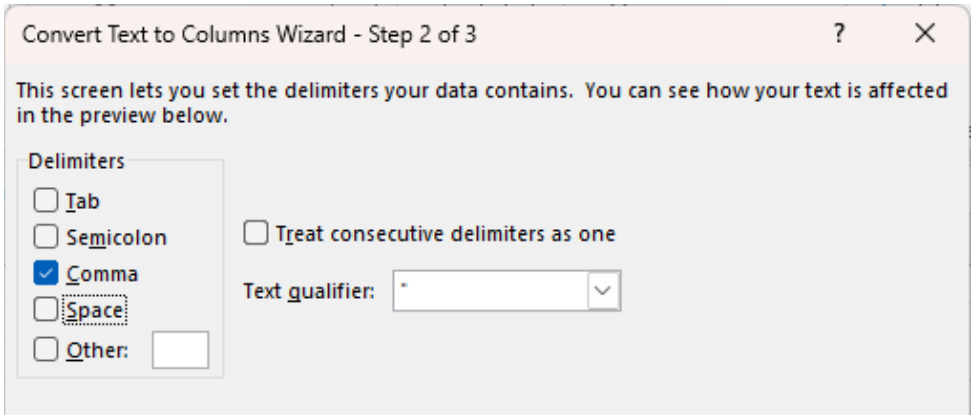
   a) Start by pasting the list of states (commas and all) into cell A6.

   b) Launch the Text to Columns wizard, and choose Delimited in Step 1. Then in Step 2, notice what happens if you try to use the delimiters for both Comma and Space:

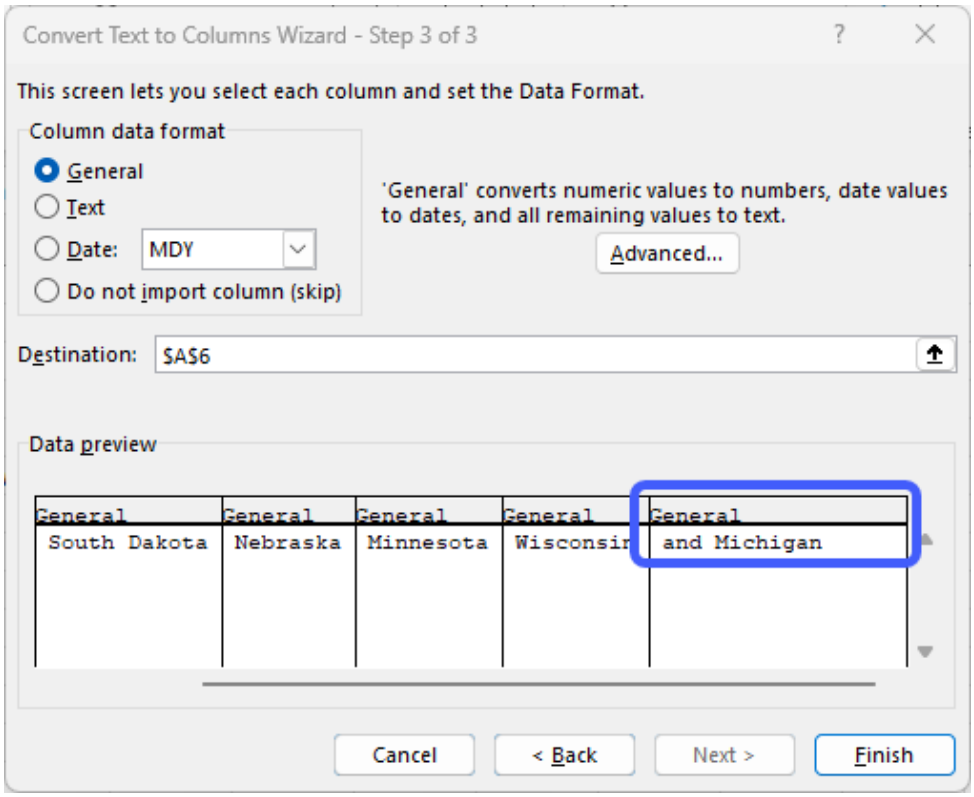


Using the space as a delimiter breaks up “North” and “Dakota”, and “South” and “Dakota” into separate columns – not what we want!

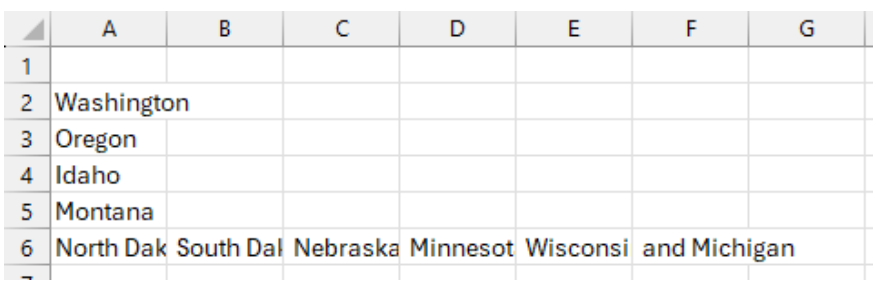
Instead, uncheck Space, and use just the comma as your delimiter, and click Next.



c) Complete Step 3, this time skipping no columns. (Notice that because we did not use the space as a delimiter, “and” is not in its own column so can’t be skipped.)



d) This leaves us with a horizontal row like this:

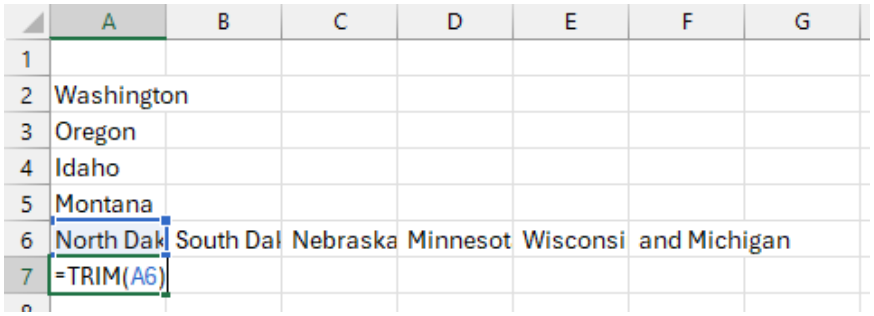


	A	B	C	D	E	F	G
1							
2	Washington						
3	Oregon						
4	Idaho						
5	Montana						
6	North Dak	South Dal	Nebraska	Minnesot	Wisconsi	and Michigan	

It’s difficult to see, but we are left with a problem to solve – the values in columns B through F are preceded by a leading space, as we were not able to eliminate those spaces by treating them as a delimiter. This will cause trouble if we try to compare these values to other parts of our spreadsheet that don’t have a leading space (because e.g. “ Nebraska” is not the same value as “Nebraska”).

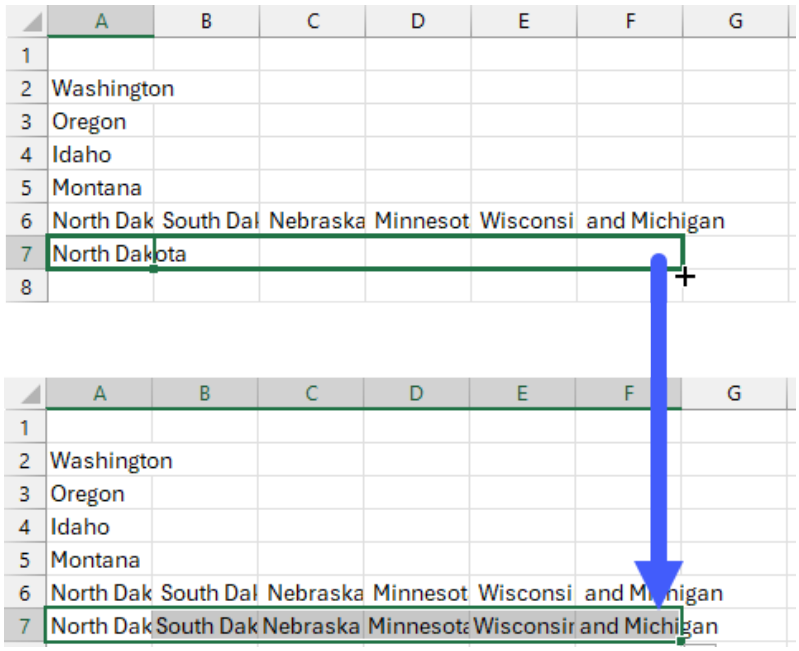
To solve this problem, we will use the =TRIM function to eliminate any pesky leading or trailing spaces in our data. Select cell A7 and enter the formula

=TRIM(A6)

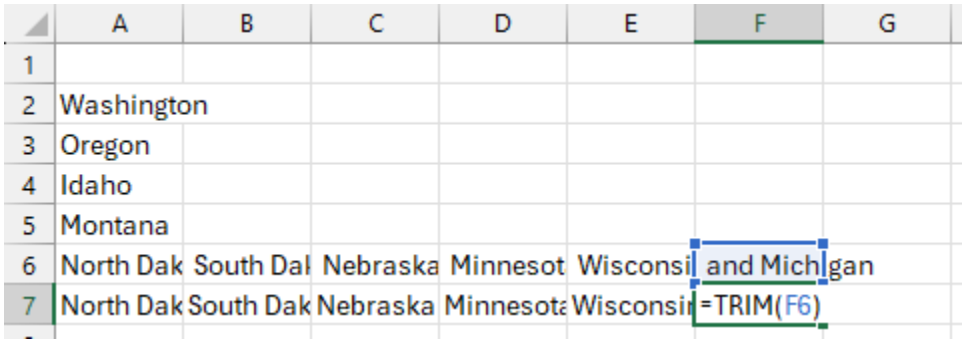


Hit enter to complete entry of the formula.

Now hover your mouse over the bottom right corner of A7 to get the + cursor. Click and drag to the right to repeat the formula for columns B through F:

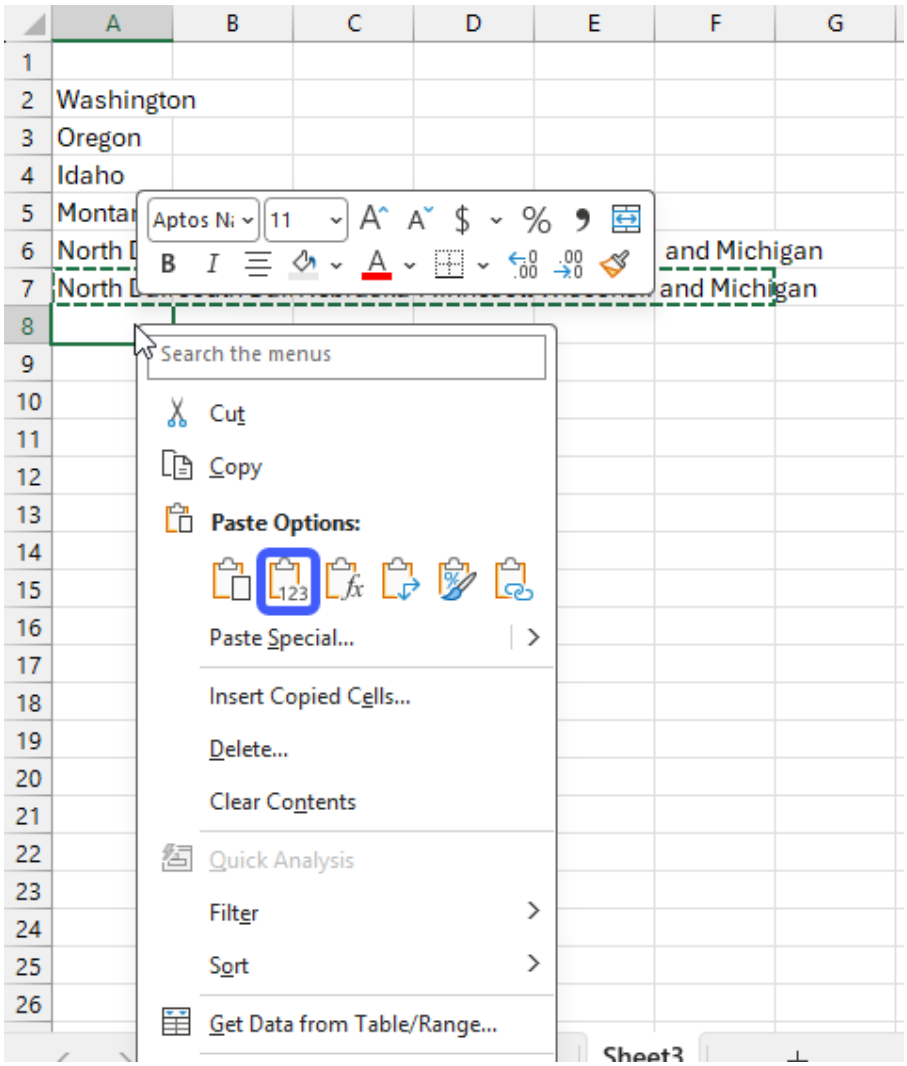


e) What happens if you try to double-click on cell F2 to manually change the value to remove “and ”?

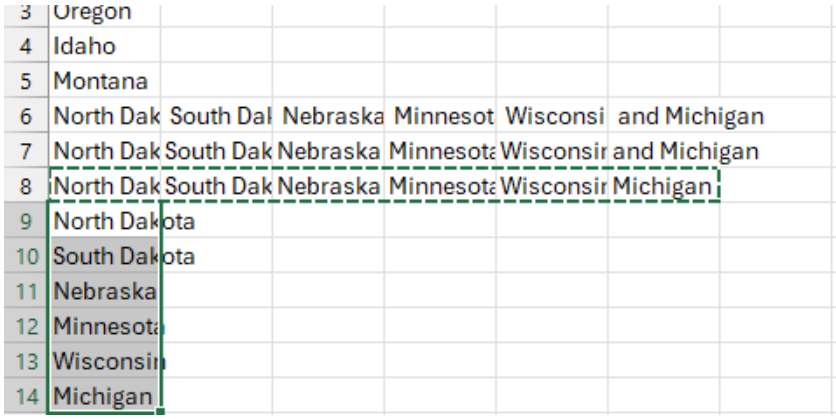


Because the value in F7 is the result of a formula, trying to edit the cell will only let you edit the formula itself. We need to convert these trimmed values to text output!

Highlight and copy A7:F7. Then right-click in cell A8, and choose Copy Values (the clipboard with the little 123 in the corner). This will copy over the cell values rather than the underlying formulas.



f) Now you have a clean set of values in row 8 – whew!! Double-click on cell F8 to remove the "and ", leaving just the state name Michigan. Then create your vertical list by copying cells A8:F8, right-clicking in cell F9, and selecting Transpose.



g) You now have the values you want, plus a bunch you no longer need. As a quick trick to help clean things up, highlight all cells A9 through F14 plus three or more additional blank rows, and hit Ctrl+C (or right-click and Copy):

	A	B	C	D	E	F	G
1							
2	Washington						
3	Oregon						
4	Idaho						
5	Montana						
6	North Dak	South Dal	Nebraska	Minnesot	Wisconsi	and Michigan	
7	North Dak	South Dak	Nebraska	Minnesota	Wisconsir	and Michigan	
8	North Dak	South Dak	Nebraska	Minnesota	Wisconsir	Michigan	
9	North Dakota						
10	South Dakota						
11	Nebraska						
12	Minnesota						
13	Wisconsin						
14	Michigan						
15							
16							
17							

Finally, select cell A6 (at the top corner of the area we want to clean up, then hit Ctrl+V (right-click and Paste). Your copied cells including the blank rows will overwrite the full area that you need to clean up. Voila!

	A	B	C	D	E	F
1						
2	Washington					
3	Oregon					
4	Idaho					
5	Montana					
6	North Dakota					
7	South Dakota					
8	Nebraska					
9	Minnesota					
10	Wisconsin					
11	Michigan					
12						
13						
14						

7. Add a header label in A1 to name the column “State”, then add another header in A2 for “Sales Region” and manually populate the appropriate values for Northwest and North Central. (Remember to make use of drag-to-fill!)

	A	B	C
1	State	Sales Region	
2	Washington	Northwest	
3	Oregon	Northwest	
4	Idaho	Northwest	
5	Montana	Northwest	
6	North Dakota	North Central	
7	South Dakota	North Central	
8	Nebraska	North Central	
9	Minnesota	North Central	
10	Wisconsin	North Central	
11	Michigan	North Central	

8. In C1, add the header “Hamburger” and in D1, add “Milkshake.”

9. In column C under Hamburger, you need to add the Google Trends data from the first tab of the spreadsheet. For this column, use the function VLOOKUP. In cell C2, enter the following formula:
=VLOOKUP('sales region lookup'!A2, 'by search term'!B:C, 2, FALSE)

   This breaks down as follows:
   Argument 1: look up the value in sheet “sales region lookup”, cell A2
   Argument 2: look for it in sheet “by search term”, range of columns B through C,
   Argument 3: look in the 2nd column of the range (2),
   Argument 4: only return an exact match (FALSE)

   Once you have entered the formula in C2, hit enter. You should retrieve the value .67:

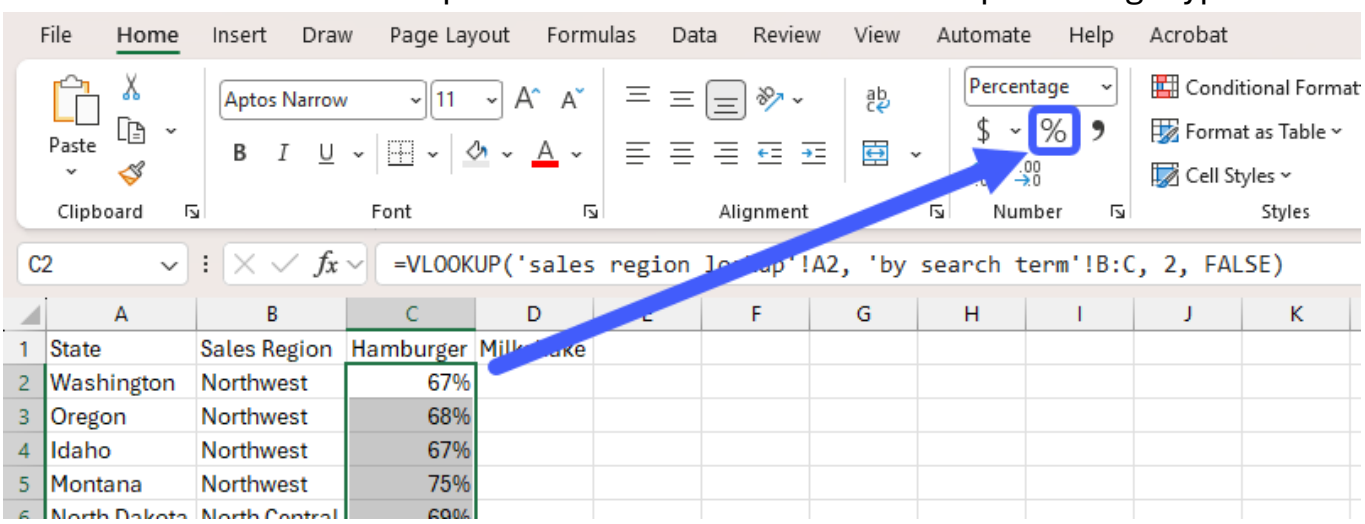
C2 | =VLOOKUP('sales region lookup'!A2, 'by search term'!B:C, 2, FALSE)

	A	B	C	D	E	F	G	H	I	J
1	State	Sales Region	Hamburger	Milkshake						
2	Washington	Northwest	0.67							
3	Oregon	Northwest								
4	Idaho	Northwest								

10. Use drag-to-fill to populate the formula for every cell down through C11:

	A	B	C	D
1	State	Sales Region	Hamburger	Milkshake
2	Washington	Northwest	0.67	
3	Oregon	Northwest	0.68	
4	Idaho	Northwest	0.67	
5	Montana	Northwest	0.75	
6	North Dakota	North Central	0.69	
7	South Dakota	North Central	0.76	
8	Nebraska	North Central	0.71	
9	Minnesota	North Central	0.67	
10	Wisconsin	North Central	0.66	
11	Michigan	North Central	0.66	

11. In the sheet "by search term," all of these values showed up formatted as percentages, but here they're appearing as decimal values. Find the % button in the Home tab of the ribbon to update the column C values to the percentage type.



12. In the Milkshake column, try using XLOOKUP instead. In D2, enter the following:

=XLOOKUP(A2, 'by search term'!B1:B52, 'by search term'!D1:D52, "not found")

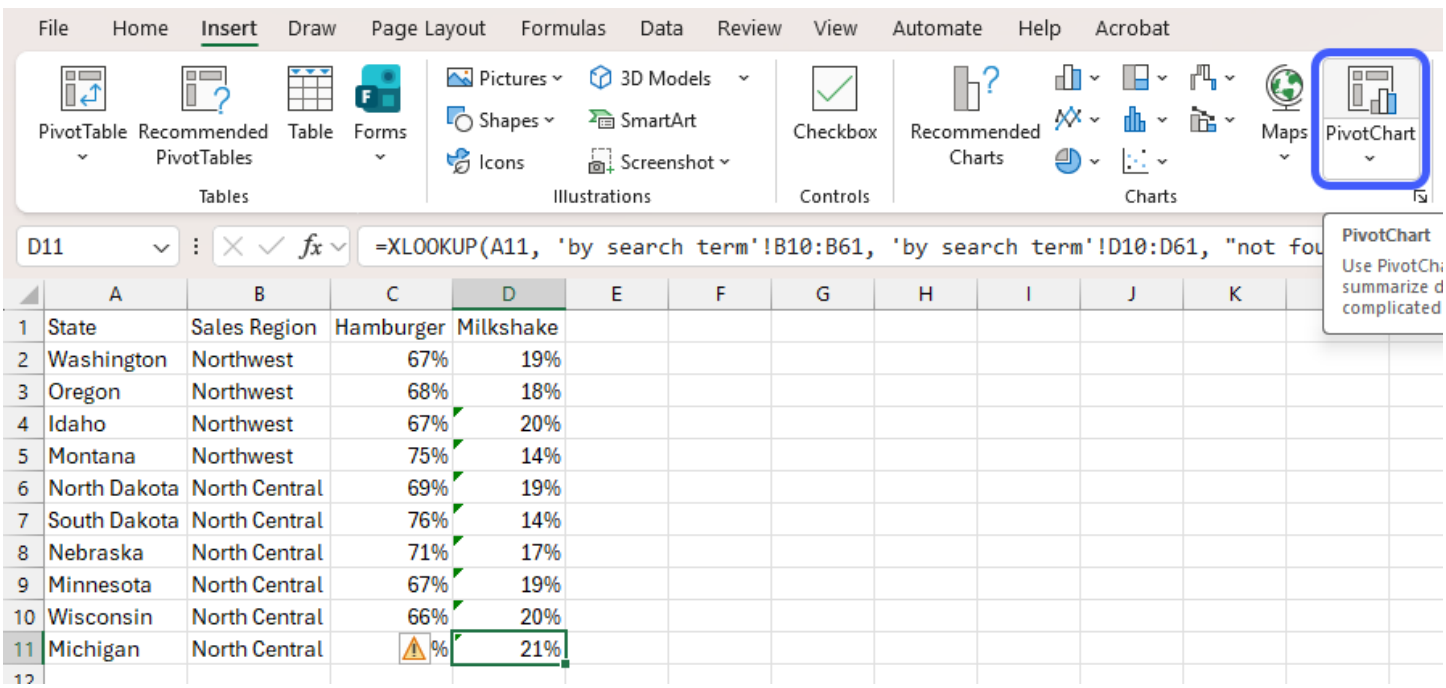
This breaks down as follows:
Argument 1: look up the value in sheet "sales region lookup", cell A2
Argument 2: look for it in sheet "by search term" in the range B1:B52
Argument 3: find the corresponding value in "by search term" range D1:D52
Argument 4: if the value is not found, return "not found"

13. Use drag-to-fill to populate the formula for every cell down through D11:

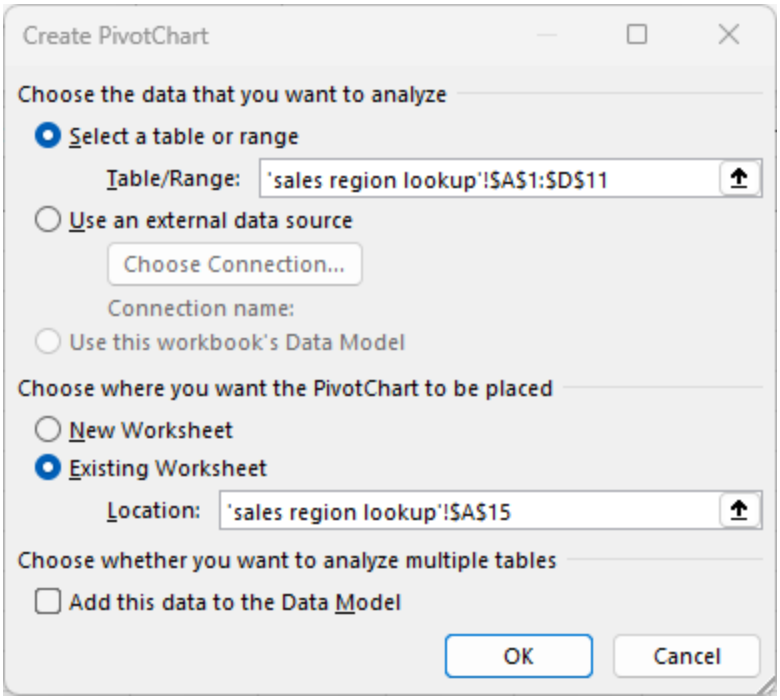
	A	B	C	D
1	State	Sales Region	Hamburger	Milkshake
2	Washington	Northwest	67%	0.19
3	Oregon	Northwest	68%	0.18
4	Idaho	Northwest	67%	0.2
5	Montana	Northwest	75%	0.14
6	North Dakota	North Central	69%	0.19
7	South Dakota	North Central	76%	0.14
8	Nebraska	North Central	71%	0.17
9	Minnesota	North Central	67%	0.19
10	Wisconsin	North Central	66%	0.2
11	Michigan	North Central	66%	0.21

Then apply the % type to the column D values.

14. Click to select the bottom right cell of your data, D11. Then go to the Insert tab, and click Insert PivotChart.

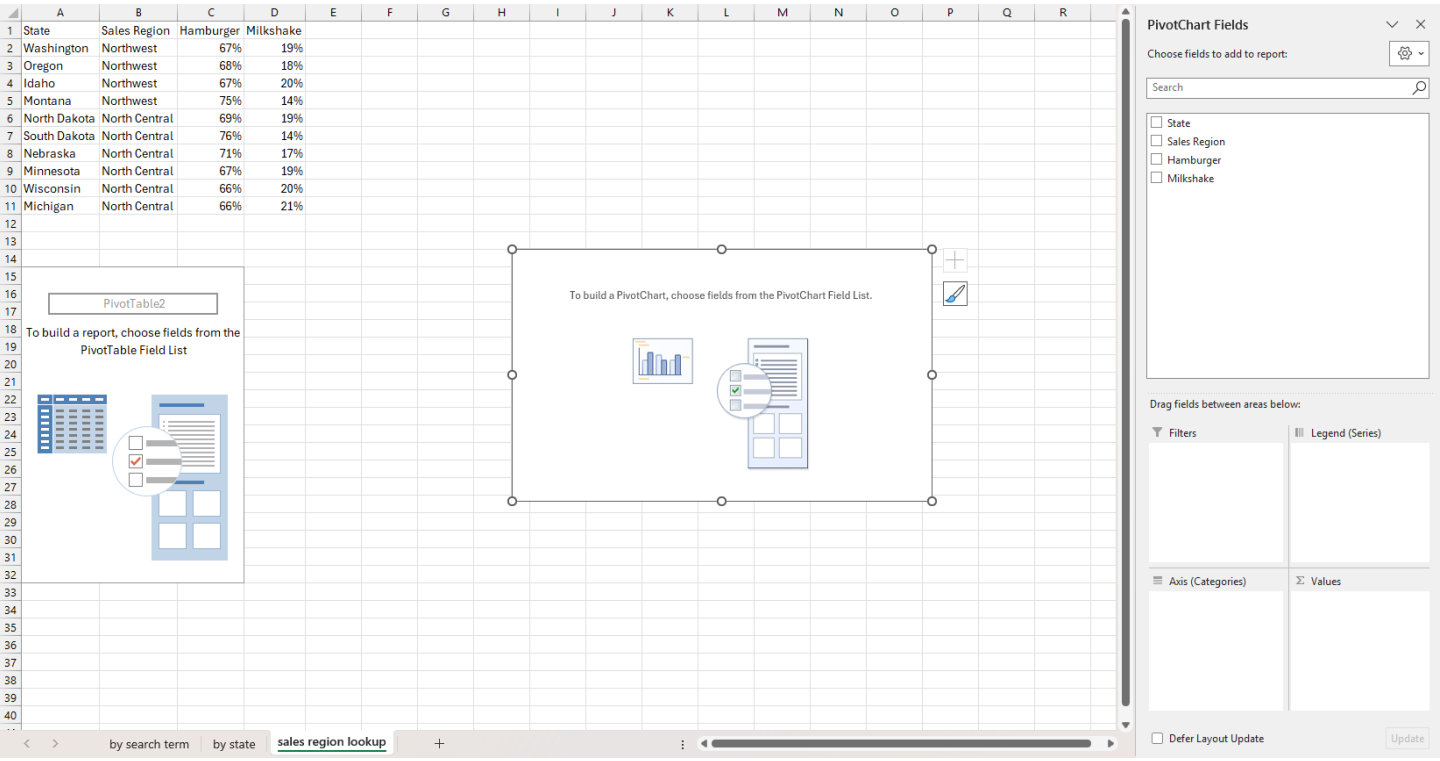


15. The Create PivotChart dialog will pop up. Under the option for “Select a table or range,” your table data should be detected as the full range A1:D11. Under “Choose where you want the PivotChart to be placed,” pick “Existing Worksheet” and click to select A15 on the “sales region lookup” sheet.



Click OK.

16. This will add a PivotTable area and blank PivotChart to your sheet, along with the PivotChart Fields panel on the right:



State	Sales Region	Hamburger	Milkshake
Washington	Northwest	67%	19%
Oregon	Northwest	68%	18%
Idaho	Northwest	67%	20%
Montana	Northwest	75%	14%
North Dakota	North Central	69%	19%
South Dakota	North Central	76%	14%
Nebraska	North Central	71%	17%
Minnesota	North Central	67%	19%
Wisconsin	North Central	66%	20%
Michigan	North Central	66%	21%

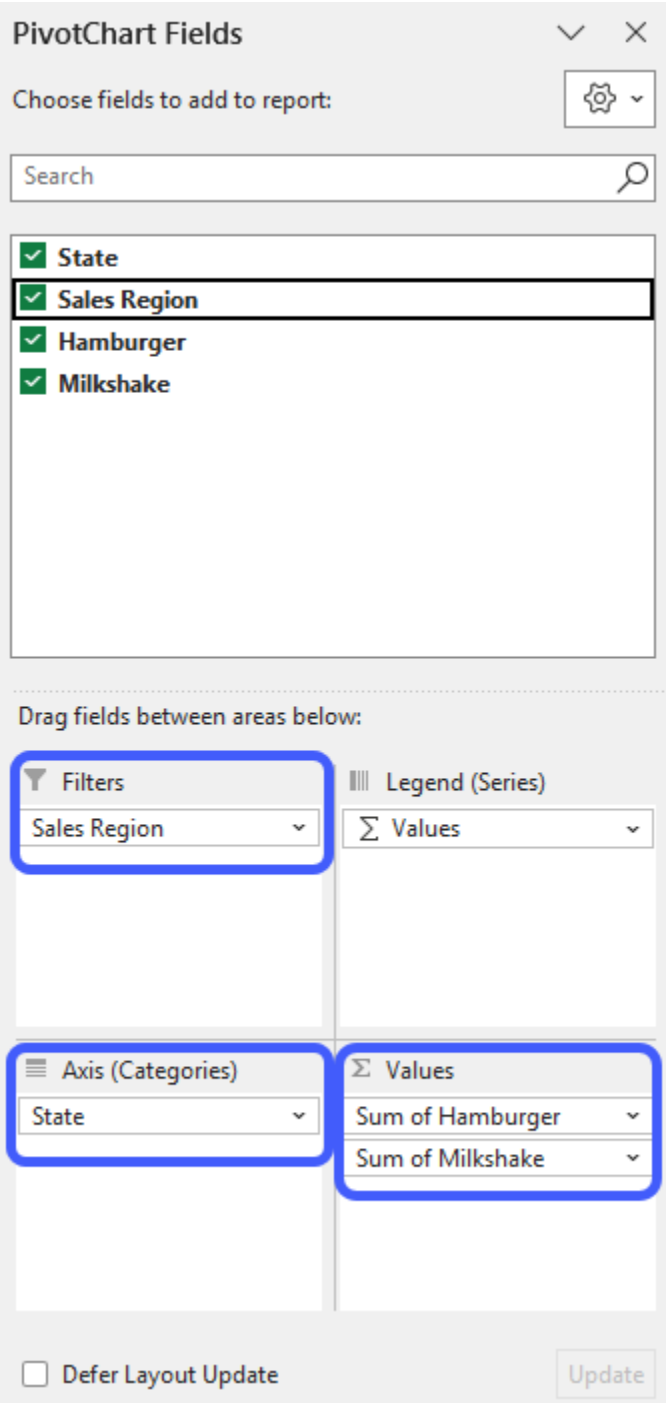
17. In the PivotChart Fields panel:

    a) Click and drag the value for State into the well for Axis (Categories).

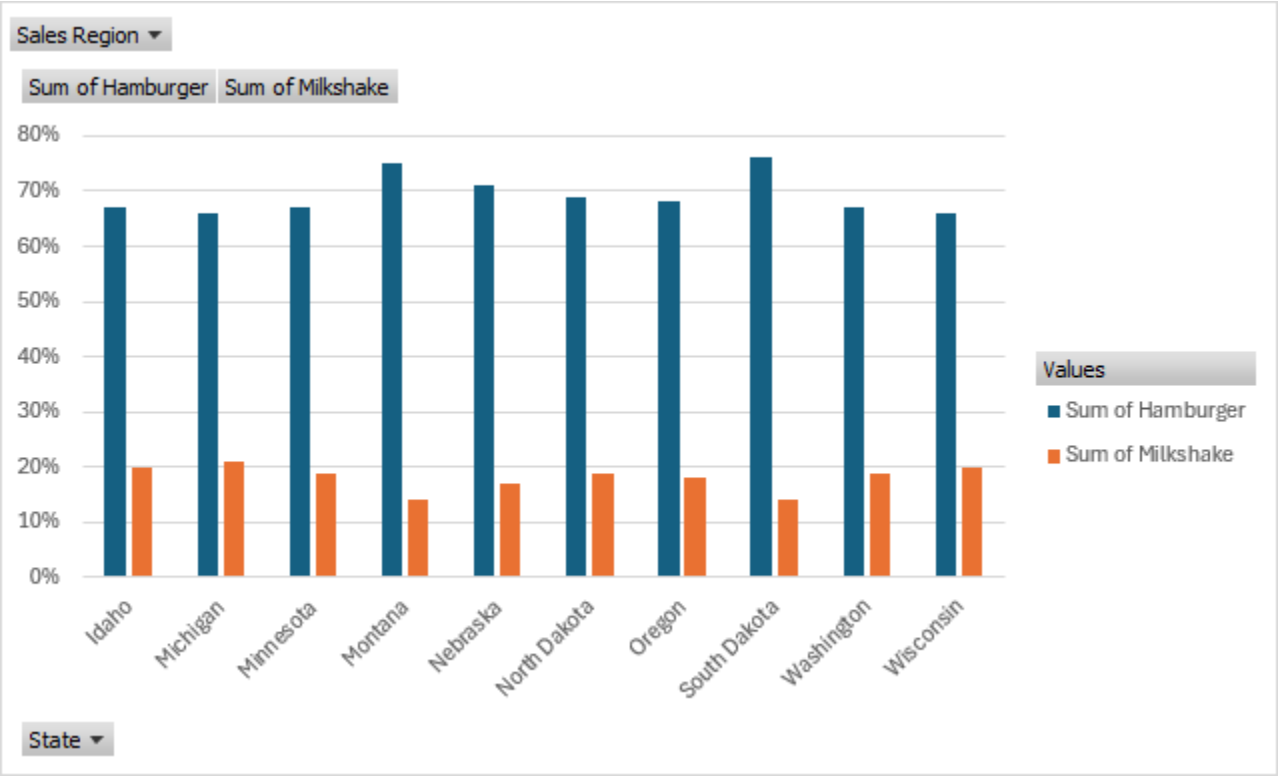
    b) Drag Hamburger and Milkshake into the well for Values.

    c) Drag Sales Region into the well for Filters.

    d) Notice that the well for Legend automatically was assigned to use Values as its source.



e) Review the chart that is created using your selections. Be sure to click on the filter for Sales region at the top to see each option:



18. Make sure your changes are saved. Then using Git Bash, commit and push your changes to the DataAnalytics repository.

Module 5

Data Storytelling

Exercise 5.A Telling data stories with comics

Getting a feel for data storytelling

1. Take 15-20 minutes to review each of these comics from the webcomic xkcd by Randall Munroe:
   a) Self-Description, https://xkcd.com/688/
   b) Lakes and Oceans, https://xkcd.com/1040/
   c) A Timeline of Earth's Average Temperature, https://xkcd.com/1732/

2. Think about the following questions:
   a) What information is being illustrated in each visualization?
   b) What do you think is interesting or engaging about each of these examples?
   c) Each example uses a different visualization format. For each of them, why do you think the author chose that format? How does the format support the information being shared?
   d) Did these examples make you think? Did they evoke any feelings from you?
   e) Which of these do you think tells the most compelling story? Why?

3. In the class discussion channel, write a brief post that identifies your favorite of the three comics and explains your pick.

4. **Bonus** (optional): Also share an example of an infographic or illustration that you found outside of class that you think is an interesting way to tell a story about data.

Exercise 5.B Restaurant customer survey results

Reviewing the results of your customer survey

1. Return to the survey you created in Exercise 3.C for your favorite restaurant using https://forms.office.com/

2. Go to View Responses, and download the survey results as an .xlsx file to your homework folder for week-01. Then open the results in Excel.

   a) Alternatively, you can open the spreadsheet using Excel online, then use the search box at the top to find the option Open in Desktop App. Then use **File** > **Save a Copy** to save to your week-01 homework folder.

3. Take a look at the columns generated by the survey.

   a) The first three columns were auto-generated by Microsoft Forms. What do each of these columns represent? Why do you think the "id" column was included?

   b) Notice how a column is assigned to each question from the survey, and each row represents one person's responses.

4. Excel uses flexible data-typing by default and formats values based on how it reads the apparent data type. Mostly this works pretty seamlessly, but occasionally it can create weird looking results.

   a) Highlight column B, **Start time**. In the **Home** tab of the ribbon, find the options for **Number**. Notice that the drop-down for number format reflects the type **General**.

   b) Use the drop-down to change the data type to **Short Date**. What happens to the column text?

   c) Now change the drop-down back to **General**. What happens?

      * Because Excel stores date values as numbers in memory, changing back to General exposes that stored number value.

      * The quickest way to get back to the original date format is simply by hitting **Undo** (ctrl+Z) a couple of times.

5. But what if we want to set a custom date format? For instance, let's say we want to include a leading zero on the month and day for single-digits (e.g. 01/01/2026) along with the precise time (e.g. 08:15:12 representing 8:15am plus 12 seconds).
   a) Use the number format drop-down to select **More Number Formats**.
   b) In the pop-up window, take a look at the available formatting categories on the left. First go to **Date** and check if the format we want already exists. What's the closest available option?
   c) Now go to the category **Custom**. If you scroll through the list of existing types, you can see that there are date formats described as **m/d/yyyy** and some variations; time formats described as **h:mm:ss** and several variations; and even an option for **m/d/yyyy h:mm**.
   d) Select **m/d/yyyy h:mm**, and check the spreadsheet again. See how it's *close* to what we want, but still missing the leading zero, and now missing seconds as well.
   e) Return to **More Number Formats** > **Custom**. In the text box above the list of types, the current type **m/d/yyyy h:mm** should be displayed. Click into this field, and edit the text to read **mm/dd/yyyy hh:mm:ss** then click OK.

6. Now that you've added this date format to the custom options, you *could* walk through similar steps to apply it to column C, **Completion time**, as well. But there's a quicker way!
   a) Click on one of the datetime values the **Start time** to select it.
   b) In the Home tab of the ribbon, find the Clipboard options, and click on the paintbrush icon for **Format Painter**.
   c) Move your mouse back into the spreadsheet data, and notice that your cursor now looks like a plus symbol with a paintbrush ( ).
   d) Click and drag down the **Completion time** column to reformat the data using Format Painter. This "paints" the format from the first cell(s) you selected into the target cells.

7. If you recall, the summary analysis by Microsoft Forms included an "average completion time" with the results page. How can you find that result from the spreadsheet data? Let's try out one way to do it.

8. Right click on column D and select **Insert** to add a blank column.

a) To create a formula to measure the amount of time taken to complete the survey by each person, first click in the first blank of the new column. Type in the formula **=C2-B2** then hit enter.

b) When you hit enter, not only will the cell that you selected populate, but Excel should auto-complete the full column. In this situation, that is exactly what we want!

* Why does Excel auto-complete the column? Because the data here is using a special format called table format. In Excel when data is "formatted as a table," that means that Excel knows exactly what the column headers are, and it can apply visual formatting like alternating row colors. It also enables some additional enhancements around how calculations are applied across the table.
* If for some reason the column does not automatically fill, check that the cell D2 where you added the formula is still selected, then use drag-to-fill down the column to fill in the formula for the remaining cells.

a) Notice that the format of your new column D looks very goofy. This is because Excel copied the datetime format used in columns B and C. Using the number format drop-down that you explored earlier, set a custom format for this column as **mm:ss**

b) Rename the column in row 1 from **Column1** to **Time to complete**.

9. Now find the average completion time. This time instead of just copying a formula from this lab workbook, use the Insert Function tool to find the right function and add it to the spreadsheet:

a) Scroll to the bottom of the new column D and select the first blank cell, after all the survey data.

b) Go to the Formulas tab of the ribbon, and click on Insert Function.

c) Find the function that lets you calculate an average (arithmetic mean), either by searching or by browsing the category for Statistical functions. Once you find the appropriate function to calculate an average, select it then click OK.

d) The pop-up wizard for Function Arguments will appear. (Recall that "arguments" are the pieces of information you pass into a function.) Take a moment to read the description of the function for your own reference.

e) In the pop-up, the top blank should default to **D2:D[number]** based on Excel's prediction logic. (If for some reason there is nothing in the top blank, use your

mouse to select the values in column D on your spreadsheet. Clicking to select may add the data as **Table(Time to Complete)**, which is also fine.) Then click OK.

f) You should now see the average of all completion times at the bottom of column D.

10. Review the columns for your survey questions. Are there any other numeric responses (like a customer age or number rating) for which you can calculate an average? Add any such calculations to the bottom of your survey results.

11. What about summarizing text data? This can get a little more tricky!

a) Pick a column where the response was multiple choice. For the purposes of an example, a multiple choice question might look something like, “How likely are you to sign up for our loyalty program?” which has three possible responses:

* Not likely
* Somewhat likely
* Extremely likely

b) Pick a response that you would like to count up. For instance, for the sample question above, how many people responded either “Somewhat likely” or “Extremely likely”?

c) Use the function **COUNTIF** to count up the number of the specific responses you want. At its simplest, if you are only counting up one specific response, your formula might look like this:
=COUNTIF(range, "Response you want to count in quotes")

To count up multiple responses, you can nest them inside a **SUM** function. For our example loyalty program question, the formula would look like this:
=SUM(COUNTIF([How likely are you to continue ordering from us in the future?], "Somewhat likely"), COUNTIF([How likely are you to continue ordering from us in the future?], "Extremely likely"))

To describe that formula more simply, it breaks down to:

```
=SUM(
   COUNTIF([Column name], “Somewhat likely”),
   COUNTIF([Column name], “Extremely likely”)
   )
```

You will need to customize the formula to your down data.

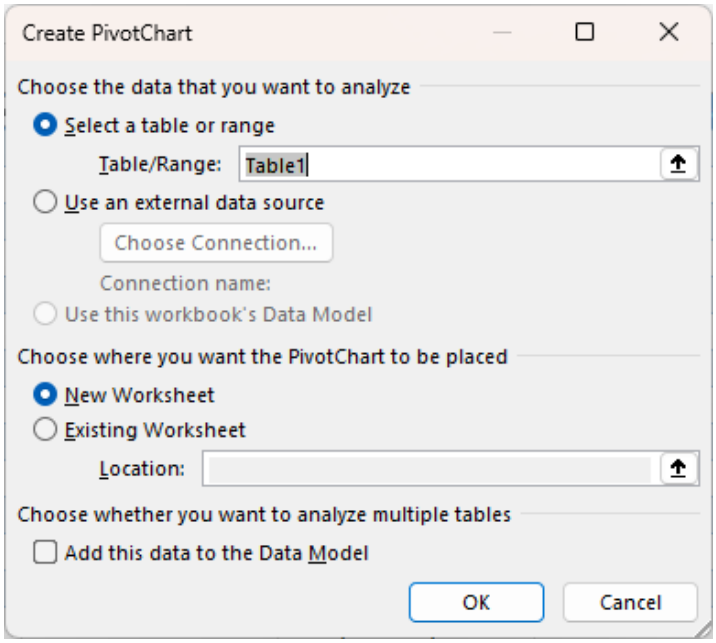
* Remember, in a formatted table, the column name is the value that appears in Row 1, and for your survey spreadsheet it will most likely be phrased as a question or sentence. If you use a column name in an Excel formula, it should always go in brackets. This *only* works if your data is *formatted as a table*.
* Instead of using the column name, you can also specify the range by cell address, e.g. M2:M22
* Text values must *always* go inside quotation marks in an Excel formula. Numbers that you are using to do mathematical calculations should *not* be in quotes.

12. Try to summarize the results for a few more survey questions using either COUNTIF or other functions.

    * For example, here's a formula that combines AVERAGE with SWITCH to convert text responses to numbers, then provide an overall average number rating: **=AVERAGE(SWITCH("Extremely dissatisfied", 0, "Dissatisfied", 1, "Neutral", 2, "Satisfied", 3, "Extremely satisfied", "4"))**
    * Also feel free to make use of other empty space in the spreadsheet to experiment with other calculations.

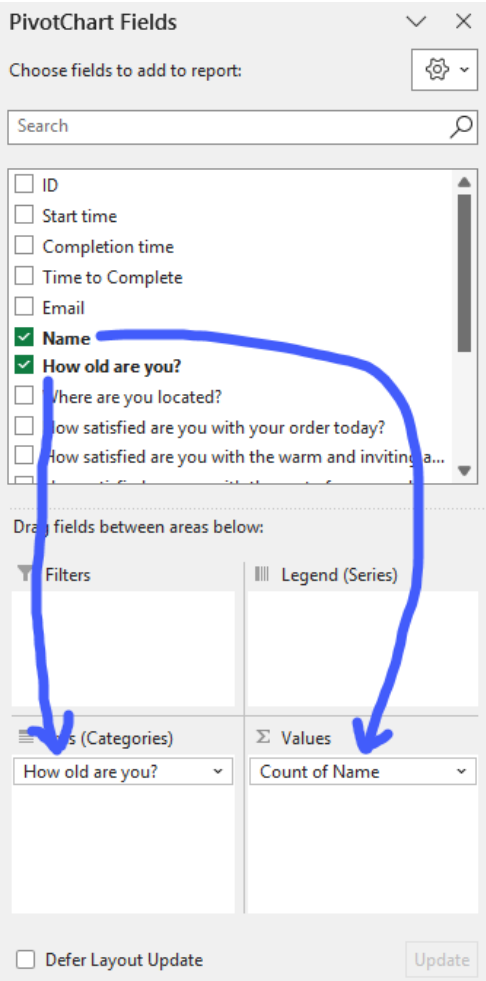
13. Try adding some PivotCharts to visualize your survey results.

    a) With any cell in your data selected, go to the Insert tab of the ribbon and click on PivotChart. Select the option to add to a New Worksheet and hit OK.

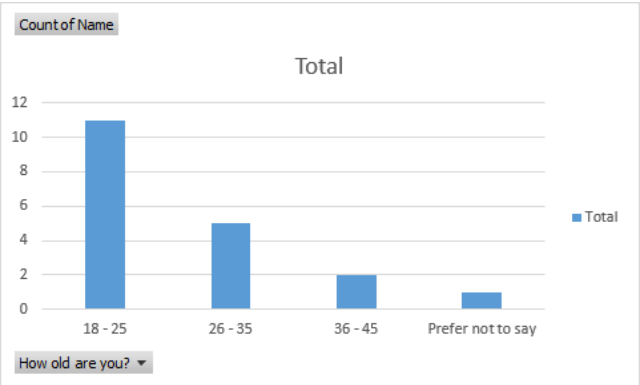


b) In the new sheet that's added, in the PivotChart Fields, try experimenting with adding a survey question to the Axis (Categories) and then adding a value to count up to the Values well.

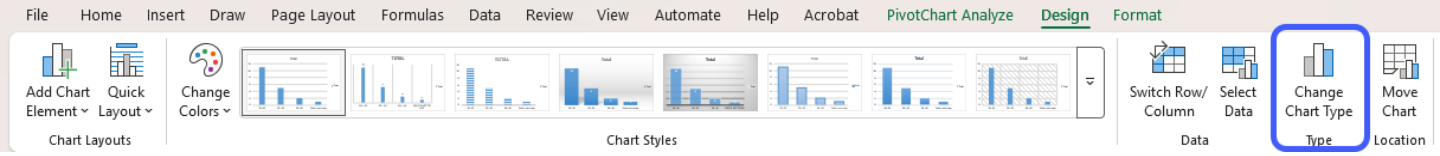
For example, if you have a multiple choice question like "How old are you" that was answered in date ranges, you can add that question to Axis (Categories). Then to Values, add a field like Name or Email that is unique to each response, to provide a count of the number of responses for each category.



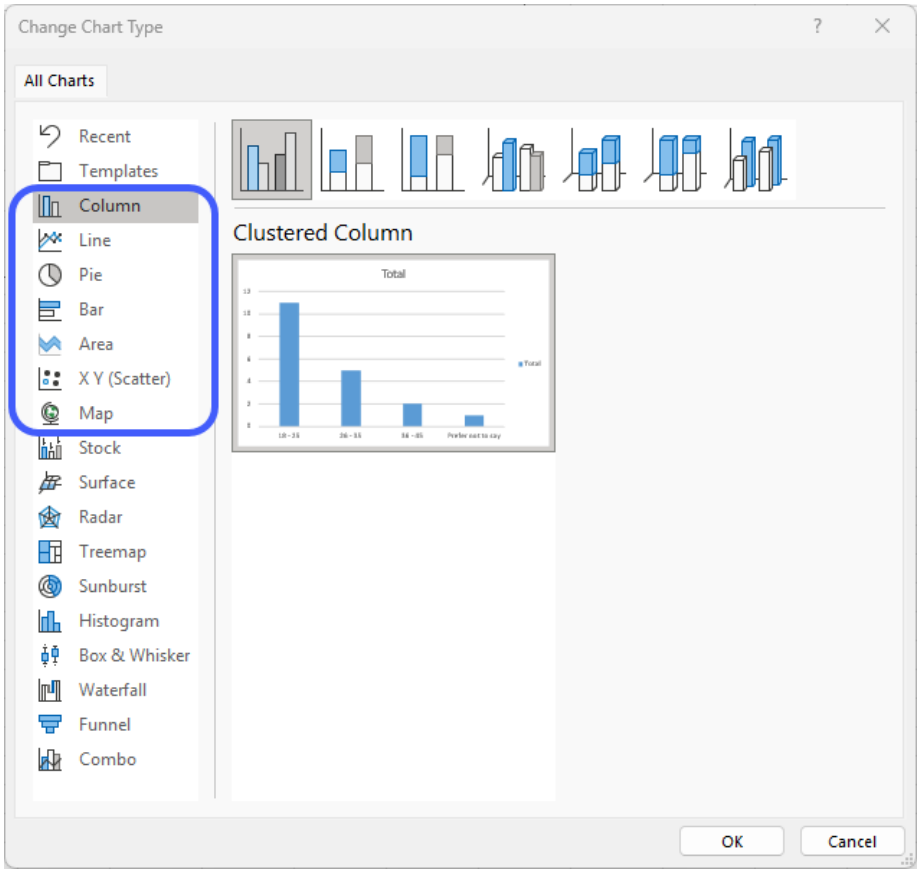
c) Excel will choose a default chart type based on the data you select:



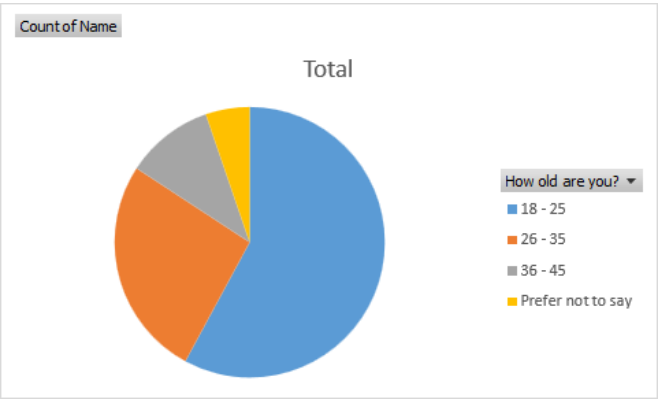
To change the chart type, click on the chart so that it is selected, then in the ribbon go to the tab for Design and click on Change Chart Type.



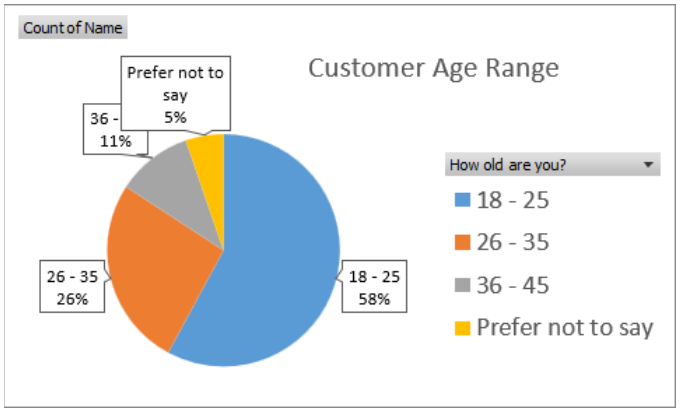
d) In the Change Chart Type pop-up dialog, explore the different chart options available.



For instance, if you are comparing number of responses by category, a pie chart might be a good option.



e) Double-click on the chart title to give it a more descriptive label. You can also resize text and change other format elements of your chart. For example:



f) If you have time, experiment with adding a second PivotChart to the same sheet, analyzing a different question.

14. Make sure your changes are saved. Then using Git Bash, commit and push your changes to the DataAnalytics repository.

End of Week 1 Lab Workbook